DIYAR: A Provenance-Graded Evidence Layer for Sustainability Claims in the Built Environment

Fatemeh Zahra Bazrgar[#1]
[#]*Faculty of Humanities and Social Sciences, Kharazmi University*
No. 43, Shahid Mofatteh Ave., Tehran, Iran
[1]github.com/Akedia-tokenize/Diyar-IBCOL2026

***Abstract*— Buildings account for roughly a third of energy-related CO_2 emissions, so climate targets depend on performance claims about individual assets being true. Often they are not: certified efficient dwellings have been measured consuming 39–54% more than their certificates predict, and a 2020 European Commission sweep judged 53.3% of environmental claims vague, misleading or unfounded. These claims now carry dated consequence: from 30 April 2026 a valuer must assess sustainability's effect on value; for older buildings the EU Taxonomy accepts top-15% performance on "adequate evidence" it does not define; and the UAE federal climate law has required emissions reporting since 30 May 2025. What is missing is not data but provenance: proof that the data received is the data the meter sent, a record of which part of a rating rests on measurement and which on an interested party's assertion, and a way to reproduce the result years later. This whitepaper designs and builds an evidence layer that grades every input by the path it arrived through, commits the record under a signed daily root for anchoring on a public chain, and makes every output independently replayable. On real, unlabelled metering data, seasonal drift pushes one-time calibration to 17.4% alerts against a nominal 5%, and recent-window calibration brings them back to about 5%; with one manipulated building among honest peers, quarterly testing reaches 80% detection only at about a 25% under-report; and a 17-mode failure taxonomy resolves at 4 tiers. The cost is about $179 per building-year under stated assumptions, nearly all of it people and physical inspection.**

***Keywords*— SDG 9, SDG 11, SDG 13, provenance grading, verifiable evidence, conformal prediction, building performance gap, bitemporal replay**

I. Introduction

Decarbonising the built environment is a first-order climate objective: buildings and construction account for roughly a third of global energy-related CO_2 emissions and about 32% of global energy demand [1]. Nearly every instrument aimed at it — retrofit subsidy, green lending, taxonomy alignment, performance standards — allocates capital on a claim about how one building performs. That claim layer is measurably unreliable: certified dwellings have been measured consuming 39–54% more than their certificates predict [2], and a 2020 European Commission assessment judged 53.3% of environmental claims vague or misleading, with up to 40% unsubstantiated [3]. When the claim is wrong, capital is misdirected and the emissions reduction is recorded rather than achieved; an evidence layer is therefore a climate instrument, not an administrative one.

The data largely exists. Dubai alone has more than 2 million smart electricity and water meters, monitored remotely every 15 minutes [4], and EU rules already set procedures for an eligible party's non-discriminatory access to metering data [5]. What is missing sits in 3 places. Nothing between meter and report proves that the figure delivered is the figure the device produced; no rating discloses how much of itself rests on measurement and how much on an interested party's word; and no published number can be recomputed a decade later, because its inputs, model and corrections were never pinned to a version.

We design and build that layer and test it on real metering data. Our contributions are: (1) a 7-level provenance grade ladder that shows, for every number, the path it came by and what it still rests on; (2) a 17-mode failure taxonomy that assigns each mode to the cheapest sufficient tier — 12 rules, 1 structural mechanism, 3 statistical tests and 1 learned model; (3) a measured detection limit: with one manipulated building among honest peers, quarterly testing reaches 80% detection only at about a 25% under-report, and less as manipulation becomes common, so below that limit provenance and random inspection carry the claim; and (4) a reference implementation on real data that carries grades end to end and replays any past output as it was known at the time.

II. Problem Statement

The obligation is dated and specific. From 30 April 2026 a RICS valuer must assess sustainability's effect on value [6]. For pre-2021 buildings the EU Taxonomy accepts top-15% performance as an alternative to EPC class A, "demonstrated by adequate evidence" it does not specify. Since 30 May 2025 the UAE federal climate law has required every entity to measure and report emissions, with its verification standard still being drafted [7]. Dubai Law No. (3) of 2026 requires a structural quality and safety certificate once a building is 20 years past completion and directs the Municipality to build a unified building database [8]; it is not an energy rule, but it builds the data infrastructure an evidence layer can connect to. Each demands evidence; none defines it (Annex F). The Green Claims Directive, meant to harmonise substantiation, has stalled since the Commission announced its intention to withdraw it in June 2025, so the gap stays open.

A. Why Existing Instruments Do Not Close It

First, the performance gap is unpredictable, not merely large: across 62 buildings the mean deviation exceeded 34% with a standard deviation of 55% [9], so no constant correction

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removes it. Second, ratings hide their composition: no valuer, lender or supervisor can see how much of a rating rests on measurement and how much on the word of the party that benefits — the exposure the 2024 standards for automated valuation models address [10]. Third, a property is a story that keeps unfolding, but a certificate is one frame of it — Dubai's new safety certificate runs for up to 10 years [8]. An evidence layer records the story, not the frame.

B. The Custodian Problem

The fourth obstacle sets the architecture: every party able to hold the record is a party to the dispute (Fig. 3).

- *Developer:* sells fast and high, and exits at handover.
- *Owner:* wants a high sale price and a low tax assessment — 2 numbers in 2 contexts.
- *Operator:* wants low maintenance costs, and supplies the self-declared fields.
- *Bank and insurer:* interested in the decisions the record feeds.
- *Certifier:* interested in its certificate staying valuable.
- *Valuer:* as custodian, would audit its own work.
- *Regulator:* incentive-neutral, but its authority stops at the border.

Neutrality cannot be promised; it has to be structural.

STAKEHOLDER	INCENTIVE TO INCREASE CLAIM	INCENTIVE TO DECREASE CLAIM	CUSTODY RISK
Developer Price at handover	▲ potential		Conflicted
Certification body Certificate reputation	▲ potential		Conflicted
Facilities operator Maintenance costs	▲ potential		Conflicted
Owner Sale price vs tax	▲ potential	▼ potential	Conflicted
Bank / insurer Decision exposure	Context-dependent	Not specified	Decision-linked
Licensed valuer Professional judgment	Context-dependent	Not specified	Self-review risk
Regulator / city Public mandate	Context-dependent	Not specified	Jurisdiction-bound
Retail investor Comparability	Context-dependent	Not specified	Limited leverage

Qualitative illustration only. Direction does not represent measured probability, impact or effect size.

Fig. 3 Trust boundaries: which way each party's interest can push the claim; only the owner pushes both ways, in two contexts.

III. Proposed Solution

A. Evidence Grading

Every field is graded by the path through which it arrived, not by the plausibility of its value; Table I states, for each grade, what the reader still relies on.

TABLE I
EVIDENCE GRADES AND RESIDUAL TRUST ASSUMPTIONS

Grade	Path	Residual assumption
A1	Device signature	Where the device is
A2	Authoritative feed	Correctness of source
A2−	Endpoint proof	The notary
B	Accredited third party	Competence, independence
B−	Unstructured document	Above, plus extraction
C+	Owner-uploaded document	Chain of custody
C	Self-declared	Everything

The ladder is a projection of a plane: who asserts a field, and what a third party needs to check it (Fig. 2). The qualified grades come from that plane — A2− is an authoritative assertion verified through a notary, C+ a third party's document whose custody runs through the beneficiary — which is why the residual assumption travels beside the grade. Grades propagate: a derived field inherits the grade of its weakest input, and an output also carries its composition — say, 60% of the energy it reports resting on A-grade evidence and 11% on self-declaration. Self-declared data is kept at the lowest grade rather than refused, because refusing it only moves it off the record, where it is ungraded and invisible.

Even A1 carries a residual assumption: a signed reading assumes the meter is where the registry says, because the signature covers the reading, not the location (Section VI).

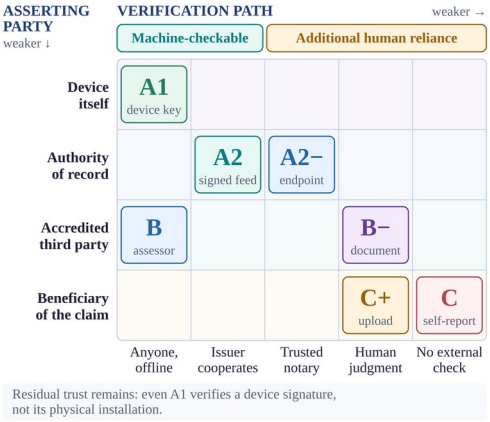


Fig. 2 Two-dimensional evidence grade space. The 7-level ladder is this plane projected onto a line; A2− and C+ exist only because the projection loses one axis.

B. System Architecture

The system has 5 layers (Fig. 1). Intake normalises meter telemetry (DLMS/COSEM [11]), operator interfaces [5], endpoint proofs, accredited assessments, bills, manual entry and authoritative feeds, and grades each field on arrival. A validation gate admits, flags or rejects. A temporal engine fits each building a baseline from its own history [12] and expresses each period as a residual, so a hot summer is not read as a failure; a baseline is accepted only if it meets ASHRAE Guideline 14 [13]. A commitment layer signs the root of the graded record daily for anchoring on a public chain, and a replay layer reconstructs any past output from committed state.

The gate has 2 parts with 2 outputs. The fast gate compares each day with the prior 28 days and emits only a *watch signal*; it opens no human case. The slow gate compares, each quarter, the building's mean residual over the last 90 days with that of its committed reference quarter and ranks this drift against peers on the same site; at $\alpha = 5\%$ it opens a *review case*. The words differ because the counts differ: about 18 watch signals per building-year against 0.189 review cases. Otherwise only a rule failure opens a human case.

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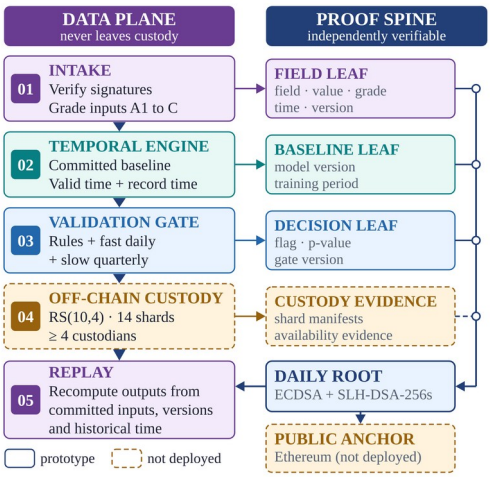


Fig. 1 The architecture as two planes. The data plane never leaves custody; the proof spine takes one leaf per stage, including the gate's own decision, and converges on one daily anchor. Dashed: not deployed.

Machine learning scores patterns in the gate and estimates unmeasured fields, recorded as estimates with their own grade; projections are emitted as intervals and never committed, since a projection recorded as fact becomes, one revision later, a fabricated measurement. Replay is bitemporal: each record carries a valid time, when the fact was true, and a record time, when the system learned it, so a late correction cannot silently rewrite history.

C. Validation Gate

We enumerated 17 ways a record can be wrong across the 7 intake paths and assigned each to the cheapest tier that suffices, because each higher tier is less auditable (Table II; Annex A gives the method, path and implementation status of each mode).

TABLE II
17 FAILURE MODES IN 4 TIERS

Tier	Failure modes
Rule (12)	Device clock drift; missing intervals; scale or unit error; stale operator cache; delivery-point ID mismatch; document forgery; post-signature tampering; wrong building or expired certificate; bill for another period or building; typing or unit slip; authoritative-feed lag; selective deletion of months
Structural (1)	Notary collusion — multi-notary committee with slashing
Statistical (3)	Copy of a prior period; optimistic rounding; physical meter relocation — drift against the committed baseline
Learned (1)	Implausible or contradictory inspector content — text model (not yet built)

No rule mode needs a model. Selective deletion is a rule, not a pattern: a deleted month cannot be found from the remaining data (E3b), so all 12 bills are required. No analysis exposes a colluding notary, so a multi-notary committee with slashing moves the assumption from one party's honesty to the cost of coordinating several. Against an adaptive machine adversary, rules and cryptography hold because their condition is structural, while statistics and models only cap its gain — which is why the slow gate's reference is a committed, immutable baseline (Section VI).

The gate turns a score into a decision by split conformal prediction [14]: flagging when the p-value is at most α bounds the probability of flagging a clean record by α, distribution-free — but under exchangeability. We name the assumption because it can fail; on seasonal data the alert rate triples unless calibration is recent (Section V).

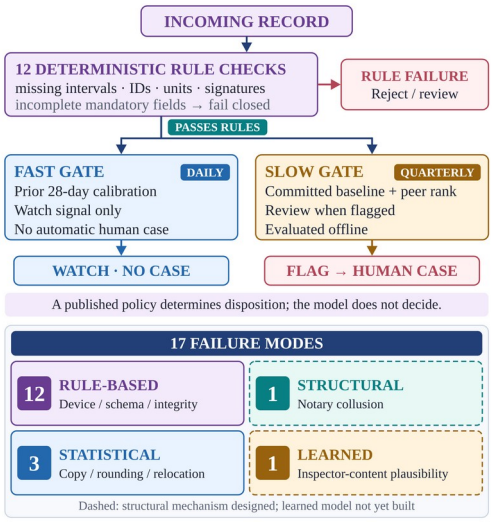


Fig. 4 Decision path through the validation gate, with the fast and slow gates. The counts below are the derivation: 12 rules, 1 structural mechanism, 3 statistical tests and 1 learned model.

IV. IMPLEMENTATION

Reference run. We built a reference implementation, DIYAR-ref, choosing each component against a stated alternative, and ran it on 3 buildings of the Fox site in BDG2 [15] from 1 January to 31 March 2017, with real readings and weather and simulated keys, through intake, grading, rules, the fast gate, signed daily commitment and replay; the slow gate was evaluated offline (Section V). It produced 91 signed days, 645 fields and 647 versions (1.74 MB of records, 5.46 MB of ledger); the gate admitted 201 items and raised 1 rule flag and 8 conformal flags. Every replay check passed: roots matched on all 91 days; January as known on 4 February reproduced the published hashes, and as known today differed only where late readings arrived (coverage 99.06% → 99.87%); a 10% edit was caught by both the meter signature and the root; one field verified alone with a 320-byte proof, and a never-entered field was proven absent. All 8 negative checks (missing weather, repeated hours, 2,000 malformed proofs) fail closed (Annex B). Authoritative claims are signed with their unit and validity and each reading is bound to its own signed installation certificate (108 targeted tests); the baselines, trained on 2016 data never

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committed as signed records, are graded C, as are the performance figures built on them. Code, data and run logs are public at github.com/Akedia-tokenize/Diyar-IBCOL2026, and akedia-tokenize.github.io/Diyar-IBCOL2026 reruns the checks in any browser.

Intake and commitment. The top grade requires DLMS Security Suite 1, not the symmetric Suite 0, which cannot give non-repudiation. The record is a sparse Merkle tree keyed by field identifier, which gives selective disclosure of one field, non-membership proofs for absent required fields, and schema extension without re-keying; KZG commitments were rejected because a break of their assumption is retroactive over a 60-year horizon (Annex E). With a 256-bit hash, Grover's search leaves about 2^{128} work for a preimage [16]; collision search already costs about 2^{128}, and known quantum algorithms do not make it cheaper once their memory is counted (Annex E).

Signatures. Each reading carries the device's ECDSA P-256 signature (64 bytes on a 43-byte record); one SLH-DSA-256s signature (FIPS 205 [17], 29,792 bytes) covers the global daily root and so protects every reading long-term, as long as SHA-256 and key custody hold (Annex E). Signing each reading with SLH-DSA-128s instead would raise 60-year storage from 11.25 TB to 837 TB.

Anchor. The design stores the daily root in an Ethereum contract's state, once per day, and the root must cite the previous one or the transaction reverts: the non-equivocation of Catena [18], enforced by contract logic, where an event-only anchor would merely make equivocation detectable. The chain is chosen for longevity, not throughput — 32 bytes a day that must verify for 60 years — and each anchor's proof bundle (transaction, receipt, receipt proof, block header) is kept in custody, as no chain promises to serve its history forever [19].

Besu. A permissioned QBFT chain orders writes once at least 4 independent consortium members exist; integrity comes from the tree and the anchor, and parameter governance, including the timelocked α, stays on Ethereum mainnet. The Dubai pilot runs on a witnessed transparency log plus the anchor; Besu enters in phase 2.

Calibration, verification and scope. Baselines are committed before the period they judge. Calibration uses a recent rolling window, not all labelled data, because one-time calibration over-alerts on seasonal data (E2); feedback updates calibration only, never the model, so reproducibility does not decay. Test vectors for 4 historical verifier failures all pass (Annex B). Keys were simulated; the anchor contract is built and tested on a local EVM, not deployed; Besu, RS custody, predicate proofs, public-randomness draws and the inspector-content model are designed, not built; the scope was 3 buildings over 3 months.

V. Evaluation

Experiments use real, unlabelled BDG2 data [15]. The main site, Fox, is the dataset's hottest (mean 25.1 °C) but cooler and drier than Dubai. Every E2–E4 result regenerates byte for byte from the 3 raw BDG2 files (Annex C).

Conformal calibration (E2). For the 106 Fox buildings that passed ASHRAE 14 (median CV(RMSE) 14.9% on the hourly 2016 fit; 13.2%, with NMBE −1.5%, on daily 2017 data out of time), baselines were fitted on 2016 and 4 calibration schemes run on March–December 2017 at α = 5%. One-time calibration flagged 16.3% of building-days and same-season (Mondrian) calibration 14.5%, because its 2016 scores are smaller than 2017's. Recency brought observed alerts back to about 5% — rolling 28-day window 5.3% (worst month 8.4%), ACI [20] 5.1% (Annex C). Without labels these are alert rates, not false positives, and ACI targets 5% by design; the one-time excess is what broken exchangeability predicts, and recency removes it. Across all 12 sites (801 buildings) the rates were 17.4%, 16.8%, 5.4% and 5.05%.

Detection limit (E3). We altered clean data from a random day onward in random buildings. With one manipulated building among honest peers, the slow gate's one-sided quarterly test at α = 5% (false-flag rate 4–5%) detects a 20% under-report with 67% power and a 30% one with 89% (Table III), and reaches 80% at about a 25% under-report (81% in a further 1,000 runs at 25%); clean quarter-to-quarter drift has a standard deviation of 13.3%. The test is one-sided in the direction the claim gains from — under-reporting for green-loan and subsidy claims (Fig. 3); a two-sided test catches the 20% under-report 9% of the time. Meter relocation (a signed meter reading a donor's rescaled load; Annex C) adds the donor's own drift: small relocations are caught more often than small under-reports, large ones slightly less.

TABLE III
Detection Power (Slow Quarterly Gate, One-Sided, α = 5%, Fox)

(a) One manipulated building, 1,000 runs per point						
Size	**5%**	**10%**	**15%**	**20%**	**30%**	**40%**
Under-report	10%	19%	38%	67%	89%	97%
Relocation	23%	33%	46%	57%	79%	92%

(b) 20% under-report by prevalence, 300 runs per point				
Share	**1%**	**5%**	**10%**	**20%**
Detected	70%	49%	32%	20%

At 1% (1 of 106 buildings) (b) repeats (a); 70% vs 67% is Monte Carlo noise.

When manipulation is common, peer comparison weakens: at 10% prevalence the 20% under-report is caught 32% of the time. Below this limit no statistical test carries the claim reliably; provenance grading and independent random inspection do. That is why evidence must be graded, not merely validated. In E3b, copying was always caught and rounding 98% of the time.

Cost. The cost of this design is human, not cryptographic. For 10,000 Dubai buildings it is about $1.79M a year, or $179 per building-year — about 12% of an ASHRAE Level 1 audit; technology is 0.5% of it, people and physical inspection 99.5%. The figure is a structured estimate: random inspection, onboarding and staff are assumptions making up 94.1% of the total, and even the review line rests on 4 assumptions. Varying inspection over 2–10% of buildings at $500–2,500 moves the figure between $139 and $379; varying onboarding too,

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between $39 and $479, with technology below 2.4% throughout. If the pilot confirms DEWA's installation registry as an A2 source, onboarding falls toward zero and the figure toward $79. Anchoring costs about $29 a year at the contract's measured gas (Annex D).

Custody. Sixty years of signed 15-minute, 5-channel meter telemetry for 10,000 buildings is 11.25 TB in one copy; the post-quantum layer adds only 0.65 GB, and the 2.49× overhead comes from device signatures. An RS(10,4) code replaces triple replication: 15.75 TB instead of 33.75 TB. The 14 shards are encrypted before encoding and spread over at least 4 independent custodians, none holding more than 4, so the system tolerates the loss or withholding of any one custodian, and the residual assumption of custody becomes the key, not the data (Annex E). A deposit prepaid at issuance, scaled to data volume, decouples storage from the operator's survival; it prices storage only, at today's prices (Annex D).

Related work. DIYAR's commitment layer descends from key transparency, which also keeps a log of map roots [21]; it adds a per-field grade with its residual assumption, valid time separate from record time, and replay of derived numbers. Against a witnessed log, the public anchor adds one permissionless checkpoint that no operator or consortium member can fork, withdraw or outlive; Annex H compares the two under equivocation, deletion and operator failure. A zkTLS endpoint proof [22] is exactly the A2− path (Annex H).

Pilot path. Three phases, each with a pass criterion; legal basis in Annex F. *Phase 0,* subject to DEWA granting data access, repeats E2 and E3 on anonymised smart-meter records, as DEWA's R&D Centre has done in a published study [23]; pass: baselines meet ASHRAE 14, the rolling gate stays below 10% each month, and Dubai's noise floor is measured. *Phase 1* shadow-runs consenting buildings without affecting decisions; pass: replay reproduces 100% of reports and onboarding cost is measured. *Phase 2* adds the consortium and qualified signatures.

VI. Discussion

The design does not make sustainability data accurate; it makes provenance legible, derivation reproducible without trusting the operator, and the grade of every input visible in the output — modest guarantees, but the ones the instruments now in force require and none supplies.

A. What This Architecture Does Not Solve

Physical binding. A signature proves which device produced a reading, not where it is. Three utility methods shrink the gap: tamper events logged by the meter and committed under its signature [24]; voltage correlation with neighbouring meters, which reveals the transformer and phase [25]; and energy balance against the transformer meter. Together they narrow the assumption to "under the same transformer, with tamper sensors disabled", on A2 utility evidence. In Dubai, DEWA's installation registry would move it to "the registry is correct" (A2), so a building-level figure would be A2 even when the reading is A1. DIYAR-ref carries each meter's signed installation certificate grade (B) into building-level energy.

Omission. A sparse Merkle tree proves absence, not intent. At enrolment DIYAR commits the list of mandatory fields before the period and the owner accepts it, and the enrolled population is reconciled against the registry; a missing field is then proven by non-membership and triggers a pre-announced consequence — dependent grades drop and a review case opens. Intent stays unproven, and the duty binds only members until a law extends it.

Consortium legitimacy. A permissioned ledger is only as neutral as its membership. In phases 0 and 1 a single operator writes, and neutrality rests on the public anchor, independent replay and witnesses.

Calibration contamination. Measured in E3: the fast gate absorbs a sustained 20% under-report, flagging it in 54% of runs on day 0 but at the clean 5.5% after day 28, as manipulated history enters its window — safe for the owner, unsafe for the lender. The slow gate's committed baseline does not absorb the shift (Table III).

Honest majority and adaptive cheats. A patient cheat below the limit, or one of many (Table III(b)), escapes statistics. Independent, score-independent random inspection is therefore not a supplement to the gate but the only mechanism that makes sub-threshold manipulation detectable — a necessary condition of the design.

VII. Ethical Considerations

The model never decides. The gate emits a flag and a record — p-value, model and calibration versions, reference peers — and a published rule turns it into an outcome. An owner must be able to contest something stated, and a rule can be argued with where a model cannot.

The product is an evidence file. DIYAR delivers the file and its grade composition; the valuer values and signs. It produces no value-adjustment factor, which would be a direct valuation input.

Neutrality is a revenue constraint. No revenue may depend on the size of the number produced; baselines are committed before any commercial relationship; disputes are resolved outside the operator; no token claims savings cash flow; and replay must run independently of the operator's servers. The first constraint is why DIYAR forgoes efficiency-financing revenue, the most profitable position in the chain.

α has a price. α trades disturbance to innocent owners against detection — a policy choice — so it is published, timelocked and changed only through governance. The peer-rank test selects αn buildings, rounded down, at every quarterly test whatever their honesty — 5 of 106 at α = 5% — so the slow gate opens 0.189 review cases per building-year, each taking 2 reviewers 1 hour at $25.54, or $9.65 per building-year. α = 1% cuts cases to 0.038 but detection of a 20% under-report from 67% to 8%. The load, 47 cases per quarter per 1,000 buildings, is set by the rule and so budgetable in advance (Annex G).

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Inspection by public randomness. Buildings for physical inspection are drawn from a public, verifiable randomness source and the draw is committed on chain, so the operator cannot choose and the owner cannot foresee.

Consent and fairness. Selective disclosure reveals one committed field, such as a lease clause, without the rest of the contract; proving a threshold without revealing the value needs a zero-knowledge range proof (designed, not built). Mondrian conditioning serves fairness across groups, not validity (E2); its wider sets for small groups are published, not hidden.

VIII. Conclusion

The obligation to evidence sustainability performance is dated, in force and unaccompanied by a method; what is missing is not data but provenance and reproducibility. This whitepaper designed, built and tested on real data an evidence layer that grades every input by its path and names the residual assumption of every grade, including the strongest. Detection has a measured limit: with one manipulated building among honest peers, quarterly testing reaches 80% detection only at about a 25% under-report, and less as manipulation spreads. Below that limit, provenance grading and independent random inspection carry the claim — which is why evidence must be graded rather than merely validated. The cost is human: about $179 per building-year under stated assumptions, nearly all of it people and physical inspection. Almost every component is adopted rather than invented; the contribution is the thin layer that does not yet exist. Its principal limitation is also its thesis: an evidence layer is worth little in its first year and more every year after, because what it accumulates is history.

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ANNEX A

Failure-Mode Taxonomy and Provenance-Grading Rules

Submitted as part of the Project

DIYAR: A Provenance-Graded Evidence Layer for Sustainability Claims in the Built Environment

Fatemeh Zahra Bazrgar

Faculty of Humanities and Social Sciences, Kharazmi University

No. 43, Shahid Mofatteh Ave., Tehran, Iran

github.com/Akedia-tokenize/Diyar-IBCOL2026

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A.1 Scope and interpretation

This annex expands Section III.C and Table II of the whitepaper. The 17 failure modes are **a design taxonomy, not 17 implemented detectors**: 7 arrival routes lead to 12 deterministic control categories, 1 structural mechanism, 3 statistical modes and 1 learned-content mode. The status of each is stated in A.3, not inferred from its place in the taxonomy. DIYAR-ref, the reference implementation, signs authoritative claims with their unit and validity, binds each meter to its building with a signed installation certificate, grades derived figures by their weakest committed input and enforces record chronology, also under optimised Python (108 targeted tests).

A provenance grade describes the **path taken by an assertion**, not the truth of the physical value. Each record carries its residual trust assumption. A derived field takes the weakest grade among the relevant inputs and can additionally report an evidence-composition breakdown. Self-declared inputs remain explicitly graded C instead of disappearing from the audit trail.

A.2 The 7 evidence grades

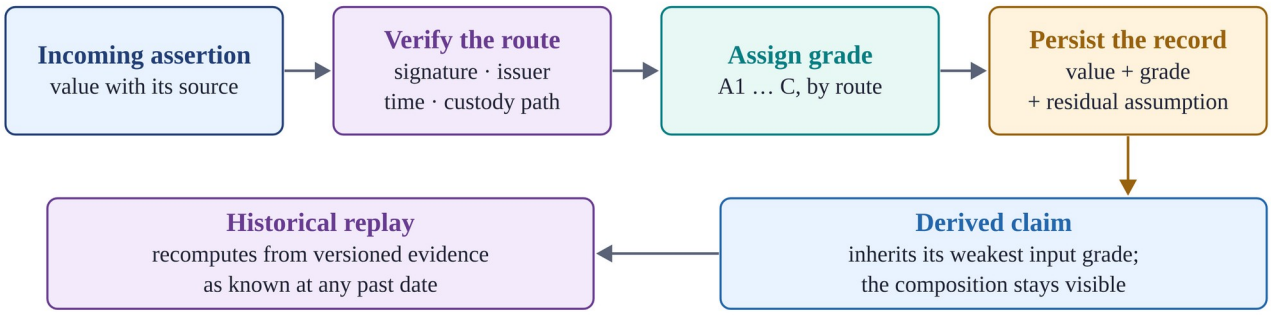


Fig. 1 Field-level provenance grading, residual trust and derived-claim propagation. The path represents design semantics; not all 7 intake adapters are implemented.

Grade	Arrival path	Residual assumption	Tested / implemented scope
A1	Device-signed telemetry	Physical device-to-premises binding	Prototype: simulated device keys
A2	Signed authoritative source	Source correctness	Prototype: signed claim with unit and validity; simulated, pinned issuer keys
A2−	Notarised endpoint proof	Notary threshold and web PKI binding	Design only
B	Accredited independent party	Competence and independence	Prototype: signed installation certificate per meter; simulated installer key
B−	Unstructured third-party document	Party competence + extraction	Design only
C+	Beneficiary-uploaded document	Chain of custody	Prototype owner upload
C	Self-declared value	All assertions and context	Prototype manual entry

Evidence: diyar/records.py, diyar/intake.py, diyar/binding.py, diyar/trust.py and whitepaper Section III.A. A2− endpoint notarisation and B− extraction from unstructured third-party documents are not implemented in the reference prototype.

A1 device signing proves that a key signed a reading, not that the device is installed at a particular building. DIYAR-ref therefore binds each meter to its building with a signed installation certificate (device key, building, channel, unit and validity window, signed by a pinned installer key) and carries the certificate's grade, B, into building-level output. A missing, revoked or relabelled certificate makes the report fail closed rather than fall back to A1; negative test N7 and the binding tests check these paths. The certificate does not establish actual premises

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occupancy or installer accreditation. An authoritative utility registry could raise the binding to A2, contingent on verified access to it and its correctness.

A.3 Full mode register

ID	Intake route	Failure mode	Tier	Auditable control	Status
01	Signed meter	Device clock drift	Deterministic	Compare authenticated time reference	Designed
02	Signed meter	Missing intervals	Deterministic	Count unique in-period timestamps; require coverage	Implemented core
03	Signed meter	Scale or unit error	Deterministic	Plausibility and unit/scale bounds	Implemented: range bounds; unit checked against signed feeds and certificates
04	Operator interface	Stale intermediate cache	Deterministic	Timestamp and sequence counter	Designed
05	Operator interface	Delivery-point identifier mismatch	Deterministic	Reconcile utility/service registry	Designed
06	Inspection certificate	Forged document or signature	Deterministic	Verify issuer signature and registry	Designed for documents; issuer signatures verified for feeds and installation certificates
07	Inspection certificate	Post-signature tampering	Deterministic	Verify document digest	Designed
08	Inspection certificate	Wrong building or expired certificate	Deterministic	Registry ID and validity-window check	Implemented for installation certificates; designed for inspection certificates
09	Uploaded bill	Other property or billing interval	Deterministic	Match property, billing dates, source	Designed
10	Manual entry	Typing or unit mistake	Deterministic	Range and order-of-magnitude check	Prototype range only
11	Authoritative feed	Delayed feed synchronisation	Deterministic	Compare source and ingestion timestamps	Designed
12	Uploaded bills	Selective omission of adverse months	Deterministic	Require all 12 bills; non-membership proof	Designed; E3b offline
13	Endpoint proof	Notary collusion	Structural	Threshold notaries / economic penalty	Designed
14	Manual entry	Copying prior period	Statistical	Exact-match rule plus near-copy similarity test	Evaluated E3b offline
15	Manual entry	Optimistic rounding	Statistical	Terminal-digit/ rounding test; degrade grade	Evaluated E3b offline
16	Signed meter	Physical device relocation	Statistical	Committed-baseline peer drift plus binding	Evaluated E3 offline
17	Inspection report	Implausible or contradictory content	Learned	Text model with human contestability	Not built

Evidence: diyar/gate.py, diyar/intake.py, diyar/binding.py, src/e3a_quarterly.py and src/e3b_statistical.py.

There are 4 boundary cases. Mode 14 includes trivial exact-copy equality but uses statistical similarity for noisy copies. Mode 15 detects a rounding pattern, not the hidden amount of favourable bias. Mode 12 requires all 12 billing months and proves a field absent from a particular commitment, but not why it was omitted. Mode 13 is not

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solved by an anomaly model: colluding notaries require threshold governance and an enforceable economic mechanism.

A.4 Decision boundary and human contestability

The implemented prototype verifies signatures and deterministic hourly coverage and range checks, then records a rolling 28-day conformal score with the calibration and model versions. The fast gate produces watch signals and does not itself open human review cases. The slow quarterly gate is a separate horizon, evaluated offline against a precommitted reference and site peers. A hard rule failure can also open a case. The gates run on parallel horizons, not as a serial fast-then-slow pipeline.

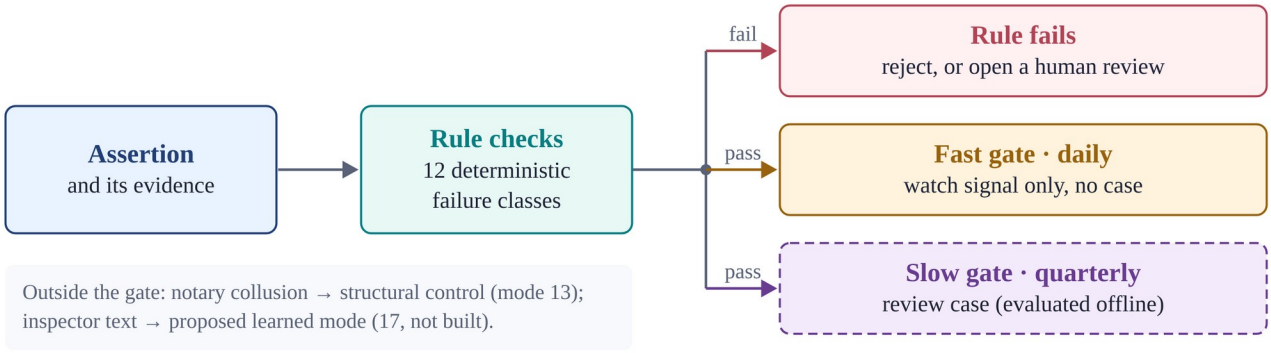


Fig. 2 Detection mechanisms and escalation boundaries. Fast and slow statistical gates have distinct schedules; the slow gate was evaluated offline.

The intended decision record identifies the affected field, grade, residual trust, source/custody evidence, rule outcomes, model and calibration versions, peer cohort, p-value when applicable, valid time, record time, and signed root. The current prototype implements only the stated subset. Adverse financial or regulatory decisions require review and contestability; a model output is evidence for that process rather than a decision-maker.

A.5 Residual adversarial risks

- **Tamper-evidence is not truth.** A compromised but valid meter key can sign incorrect readings; physical and regulatory controls are needed.
- **Absence is not intention.** SMT non-membership cannot prove that a missing bill was deliberately suppressed.
- **Peer reference contamination is measurable.** E3 shows under-report detection declines when manipulation becomes prevalent.
- **Notary thresholding relocates trust.** The number of notaries, selection, penalty and dispute process are not implemented here.
- **Learned inspector-content assessment is not built.** The proposed text model cannot be presented as a validated detector.

A.6 Completion criterion

A claim of operational coverage for all 17 modes would require independent positive and negative test vectors for every control path. The reference implementation supports a narrower statement: signed-input controls (including unit, validity and installation binding) and replay are implemented; the fast gate is implemented; the slow gate and the E3b detectors are evaluated offline; the remaining controls are designed. The whitepaper keeps this distinction.

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ANNEX B

Reference Implementation, Replay, Security Tests, and Reproduction

Submitted as part of the Project

DIYAR: A Provenance-Graded Evidence Layer for Sustainability Claims in the Built Environment

Fatemeh Zahra Bazrgar

Faculty of Humanities and Social Sciences, Kharazmi University

No. 43, Shahid Mofatteh Ave., Tehran, Iran

github.com/Akedia-tokenize/Diyar-IBCOL2026

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B.1 Repository and execution prerequisites

The evidence package is DIYAR-ref: Python source, tests, output JSON and CSV, a source-hash manifest and the cost workbook. It is public at https://github.com/Akedia-tokenize/Diyar-IBCOL2026 (folder reference/, with this whitepaper in paper/), and the website https://akedia-tokenize.github.io/Diyar-IBCOL2026/ reruns the signature, root and proof checks and the anchor contract in the browser, on the published files. The package does not contain the upstream BDG2 metering data; scripts/get_bdg2.sh obtains the 3 upstream files and checks their SHA-256 values against data_manifest.sha256. The runs reported here used Python 3.13.18, numpy 2.5.3, pandas 3.0.5, cryptography 50.0.1 and slh-dsa 0.2.5 (import slhdsa).

The reference run reported in the whitepaper was produced on 6 November 2026 from the 3 raw BDG2 files (SHA-256 match) in 312 s: 91 signed days, 645 fields and 647 versions. Its January reports are identical in every number to those of an earlier run (2 November), which is kept unchanged as an archive; only the baseline and performance grades differ (C, because the 2016 training data are not committed as signed records). The current code also verifies that archive (5 vectors, 8 negative tests, 8 replay checks), and all 91 of its daily ECDSA and SLH-DSA signatures verify with slh-dsa and with OpenSSL 3.5.5. The 108 targeted tests of signed claims, installation binding, grade propagation and record chronology pass in ordinary and optimised Python (python -O).

The signed readings are real BDG2 hourly consumption data; device, issuer and installer keys are simulated, and their pins are generated by the same run (files ending in _pins_DEMO_NOT_TRUSTED.json). The run is not evidence of a live utility integration, real custody, a deployed smart contract or approved signer credentials.

B.2 Module-to-claim matrix

Source file	Function	Status
diyar/intake.py	Signature, unit and validity checks; grade assignment	Implemented for A1, A2, C+, C
diyar/records.py	Append-only versions, as-of and two-axis query; chronology checks	Implemented
diyar/baseline.py	2016-fit per-building energy baseline / G14 screen	Implemented
diyar/gate.py	Rule checks and 28-day rolling p-value	Implemented fast gate
diyar/smt.py	Depth-256 sparse Merkle inclusion/non-membership	Implemented and fuzz-tested
diyar/ledger.py	Daily chained ECDSA/SLH-DSA root signing	Implemented with simulated keys
diyar/replay.py	Reconstruct published outputs from files	Implemented
diyar/binding.py	Signed installation certificates and revocations; fail-closed binding	Implemented with simulated installer key
diyar/trust.py	Grade of each derived figure from its committed inputs	Implemented
src/e3a_quarterly.py	Quarterly peer-rank slow gate	Offline evaluated, not pipeline
src/e4_review_budget.py	Alpha / case-budget sweep	Offline evaluated
src/reconcile.py	Reconcile reported metrics with producers	Executed; output in reconciliation.csv

Evidence: repository README and the matching source files. The quarterly slow gate runs in offline evaluation, not inside the 3-building run_pipeline.py daily execution; that distinction matters for reading “built” versus “evaluated” claims.

B.3 The 3-building experiment

The 3 buildings analysed at Fox, from 1 January to 31 March 2017, are Fox_office_Joy, Fox_education_Wendell and Fox_assembly_Renna. Their models are accepted on 2016 meter and weather data; a fourth candidate, Fox_assembly_Lakeisha, was excluded for CV(RMSE) 35.6%. Initial records on 31 December 2016 include signed

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installation certificates (B, simulated installer key), signed authoritative floor areas (A2), the committed baseline, an owner-uploaded occupancy schedule (C+) and manually entered occupants (C, for 2 buildings).

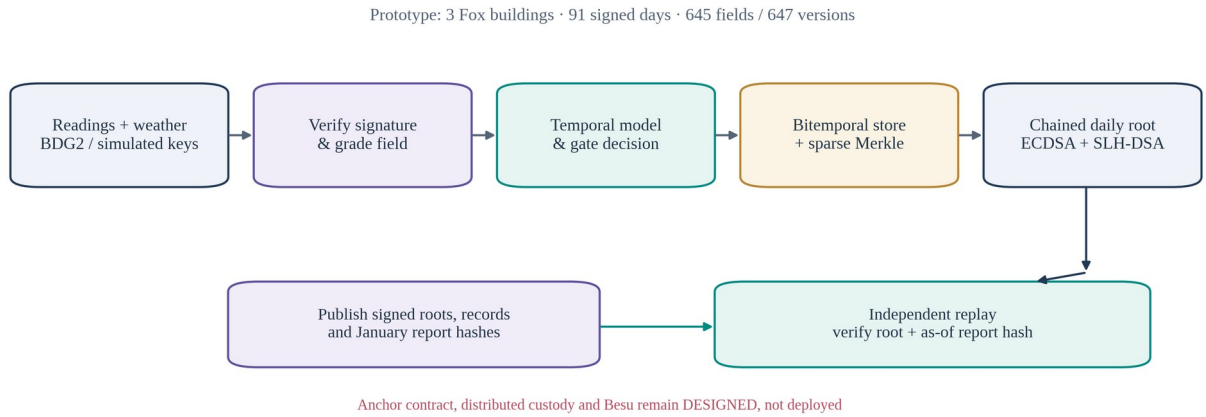


Fig. 1 Implemented reference workflow, data publication and independently reproducible replay. Blue/purple stages run locally; external anchoring and consortium are outside prototype scope.

A 6-hour outage affected Joy on 15 January. January report hashes were published on 5 February against the 4 February committed state. The missing hours arrived, signed, on 20 February with a 15 January valid time and 20 February record time. No prior version was overwritten.

Statistic	Result	Evidence producer
Signed days	91	results/v2_run/run_stats.json
Fields / versions	645 / 647	results/v2_run/run_stats.json
Record storage	1.74 MB (1,741,013 bytes)	results/v2_run/records.jsonl
Ledger storage	5.46 MB (5,459,820 bytes)	results/v2_run/ledger.json
Gate pass / rule flag / conformal flag / calibrating	201 / 1 / 8 / 60	results/v2_run/run_stats.json
Long-term daily signature	SLH-DSA SHA2-256s, 29,792 bytes	results/slh_sizes.json
Observed compressed proof	320 bytes, 9 non-default siblings	results/v2_run/replay_checks.json

The large cryptographic signature cost in a 3-building test is not representative of marginal cost at 10,000 buildings: design S4 uses a single post-quantum signature on the **global daily root**.

B.4 Replay checks

Replay of the reference run (tests/test_replay.py with the run's issuer and installer pins) verifies ECDSA and SLH-DSA on all 91 days, identical recomputed and signed roots, and valid A1 and A2 evidence signatures. Replay of January as known on 4 February matches every published report hash. Replay with the current record differs only for Joy, whose 6 missing hours were later corrected. A 10% alteration to 1 committed reading fails at both the origin signature and the state root. A single field verifies alone against the signed root (320-byte proof), and a never-entered occupants field is proved absent.

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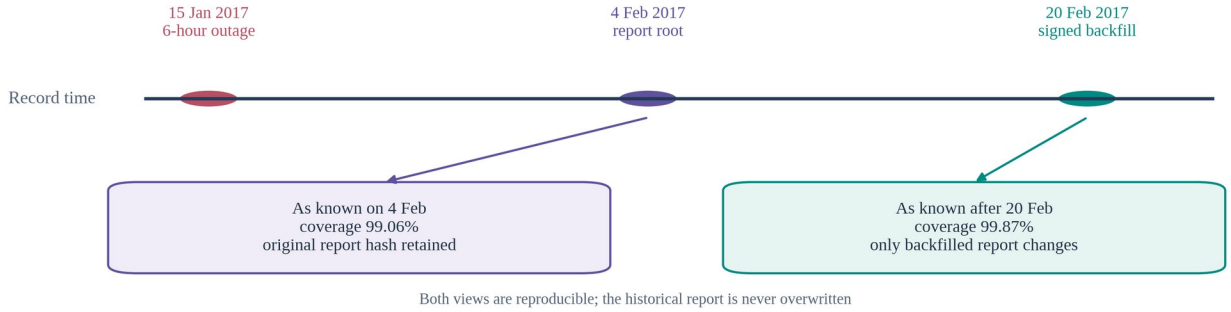


Fig. 2 Bitemporal replay of late-signed meter data: a correction changes current knowledge without rewriting the previously published January report.

Joy's recorded coverage changed from 99.059% to 99.866% (published as 99.06% to 99.87%); consumed energy from 112,692.56 to 113,461.14 kWh. This is a valid-time/record-time demonstration, not physical location verification.

B.5 Negative tests

ID	Adversarial / failure check	Result
N1	missing weather	PASS: status complete → partial; weather coverage 0.96774
N2	6 metered hours removed	PASS: drop 0.00806
N3	24 identical timestamps	PASS: coverage_ok false; 1 distinct hour, 23 duplicate readings
N4	readings stamped on another day do not count	PASS: out_of_period ×24; insufficient_coverage 0/24
N5	2,000 malformed proofs: zero crashes, zero false accepts, genuine proofs verify	PASS: crashes 0, false accepts 0
N6	bitemporal query answers both axes	PASS: got [100, 90, 100, 95, 90, None]
N7	building-level grade follows the signed installation certificate	PASS: B; refused when the certificate is removed or relabelled
N8	canonical encoding refuses NaN and Infinity	PASS: 3/3 refused

Evidence: tests/test_negative_v2.py and results/v2_run/negative_checks_v2.json for the reference run; tests/test_negative.py and results/prototype/negative_checks.json for the archive. All 8 tests pass on both, in ordinary and optimised Python.

N1 leaves metered energy unchanged when weather is missing and marks the status partial. N3 and N4 count unique in-period hourly intervals, not copied rows. N5 uses 2,000 malformed proof inputs and records zero exceptions and zero false accepts. N6 tests both bitemporal axes. N7 carries the B grade of the signed installation certificate into the building-level grade and refuses the report when the certificate is removed or relabelled. N8 rejects NaN and Infinity in canonical output.

B.6 Verifier-vector attacks

The tests/test_vectors.py suite models 4 historical failure categories: ambiguity from unseparated leaf and node hashes; a duplicate-leaf family analogous to CVE-2012-2459; malformed both-children proof use (BNB Bridge analogy); and an unbounded leaf index (Hyperbridge analogy). All pass against the project's verifier. This validates 4 locally specified adversarial vectors; it does not replicate the complete exploits in their original systems.

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B.7 Reproduction commands

```
python -m pip install numpy pandas cryptography slh-dsa==0.2.5
sh scripts/get_bdg2.sh data/bdg2
# reference run, replay and negative tests
python3 -I run_pipeline.py data/bdg2 results/v2_run
P=results/v2_run
python3 -I tests/test_replay.py $P --issuer-pins $P/issuer_pins_DEMO_NOT_TRUSTED.json \
    --installer-pins $P/installer_pins_DEMO_NOT_TRUSTED.json
python3 -I tests/test_negative_v2.py $P --installer-pins $P/installer_pins_DEMO_NOT_TRUSTED.json
python3 -I tests/test_vectors.py
for t in authoritative_v2 binding_v2 trust_p003 store_chronology_p004; do
    python3 -I tests/test_$t.py; python3 -I -O tests/test_$t.py; done
# archived run of 2 November
python3 -I tests/test_replay.py results/prototype --allow-archived-v1
python3 -I tests/test_negative.py results/prototype
# experiments E2-E4 (Annex C) into out/
python3 -I src/e2_baselines.py data/bdg2 out
python3 -I src/e2_conformal.py out/e2_daily_scores.csv.gz out Fox
python3 -I src/e3_prepare.py data/bdg2 Fox out/e3_fox_daily.npz
python3 -I src/e2_out_of_time.py out/e3_fox_daily.npz out
python3 -I src/e3a_gates.py out/e3_fox_daily.npz out
python3 -I src/e3a_quarterly.py out/e3_fox_daily.npz out
python3 -I src/e3a_extra.py out/e3_fox_daily.npz out/e3a_extra.json
python3 -I src/e3b_statistical.py out/e3_fox_daily.npz out/e3a_quarterly.json out
python3 -I src/e4_review_budget.py out/e3_fox_daily.npz out
# anchor contract: compile, then its 17 tests on a local EVM
cd anchor && npm ci && node compile.cjs && node test.mjs ../results/v2_run/ledger.json && cd ..
# every number printed in the whitepaper against its producer
python3 -I src/reconcile.py results DIYAR_cost_model.xlsx reconciliation.csv \
    --paper paper/DIYAR-whitepaper-v2.7.md --bdg2 data/bdg2
```

Evidence: repository README, SHA256SUMS and the logs in logs_2026-11-06/ and anchor/test-log.txt. reconciliation.csv checks 112 numbers printed in the whitepaper against the files that produce them (0 mismatches), confirms that each number's wording appears in the paper text (paper/DIYAR-whitepaper-v2.7.md) and lists 9 cited numbers; it tests consistency with the producers, not the external source facts.

B.8 Explicit deployment boundary

- Real meters and utility-certified signing keys: **not connected**. Physical authenticity is assumed in the source dataset; device, issuer and installer identities are simulated.
- Slow quarterly gate: **evaluated offline**, not integrated into the daily end-to-end pipeline.
- All 17 controls of Annex A: **not fully implemented**; a subset runs.
- Ethereum anchor contract: built and tested, not deployed. anchor/DiyarAnchor.sol passes 17 tests on a local EVM against the reference ledger, including rejection of a fork, a repeated day and a second writer; no root is posted on a public network.
- Besu/QBFT, multi-custodian RS(10,4), threshold key shares, zk predicate proofs and learned inspector analysis: designed, not deployed.
- JSON encoding: deterministic within Python, but no cross-language RFC 8785/JCS test-vector specification yet.
- Missing device sequence counters and compromised-but-valid signers remain separate threat categories.

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ANNEX C

Empirical Evaluation: E2, E3, E3b, and E4

Submitted as part of the Project

DIYAR: A Provenance-Graded Evidence Layer for Sustainability Claims in the Built Environment

Fatemeh Zahra Bazrgar

Faculty of Humanities and Social Sciences, Kharazmi University

No. 43, Shahid Mofatteh Ave., Tehran, Iran

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C.1 Data, design and measurement limits

This annex documents E2, E3, E3b and E4 from the outputs in DIYAR-ref (results/). Building Data Genome Project 2 (BDG2) contains hourly meter readings for non-residential buildings (2016–2017). Fox was selected as the warmest site in the dataset (reported mean 25.1 °C), but it is not a stand-in for Dubai's temperature, humidity or tenant distribution. The data carry no fraud labels: a statistical flag is not a demonstrated fraudulent event, and observed flag rates are alert rates, not verified false-positive rates.

The study covers 3 test domains. E2 measures the stability of statistical flagging under seasonal shift; E3 injects controlled manipulations into real series and estimates detection power; E4 translates a chosen review threshold into human case load. E3 and E4 are offline and not part of the daily 3-building reference run. Every output in this annex was regenerated from the 3 raw BDG2 files on 6 November 2026 and matches the published results (C.9).

C.2 E2: observed alert instability under seasonality

The 2016 source screening began with **1,578 candidate buildings**. The logged exclusions were 526 with insufficient data and 248 failing the stated Guideline 14 screen, leaving **804** accepted baselines. Only **801** of those met the additional minimum of 40 valid early-2017 evaluation days; that is why the 12-site E2 sample is 801 rather than 804. The focus site Fox contributes 106 accepted buildings and 32,323 evaluated building-days. Out-of-sample predictions use baselines fitted on 2016; comparison runs March through December 2017 at alpha 5%.

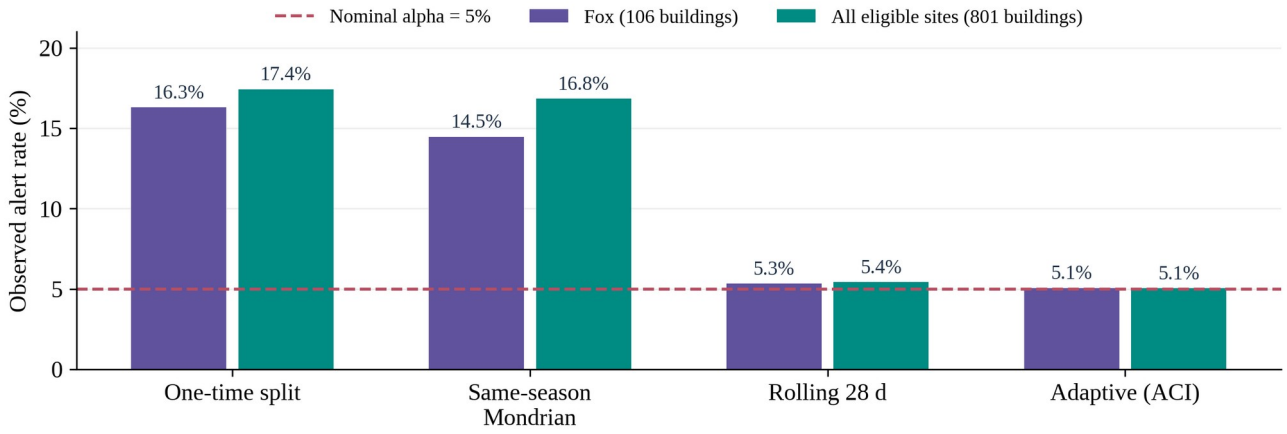


Fig. 1 Observed alert rates under 4 calibration procedures. These are alert frequencies, not empirically verified false-positive rates.

The comparison covers 4 calibration procedures: one-time split calibration on the early-2017 period; same-season (Mondrian) calibration; a rolling prior-28-day window; and adaptive conformal inference (ACI, gamma 0.005). The observed flag rates were:

Calibration	Fox rate	Fox worst month	Fox buildings >10%	All-site rate
One-time split	16.3%	21.5%	49.1%	17.4%
Same-season Mondrian	14.5%	21.1%	45.3%	16.8%
Rolling 28 d	5.3%	8.4%	0.9%	5.4%
Adaptive conformal (ACI)	5.1%	7.6%	0.0%	5.05%

Evidence: results/e2_baseline_stats.json, results/e2_summary.json, src/e2_baselines.py, and src/e2_conformal.py.

One-time and same-season calibration generated much higher observed alert loads than the nominal 5%. Recent-window calibration yielded a near-5% observed rate, and the ACI algorithm adapts its threshold in response to previous flags. In the absence of ground-truth labels, none of these is a verified false-positive rate or a direct demonstration of conformal coverage. Seasonal drift can threaten score exchangeability, but the experiment does not constitute a formal exchangeability test. Any claim of statistical validity must retain its assumptions explicitly.

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The 28-day window has a discrete rank resolution: the minimum empirical conformal p-value is 1/(28+1) = 3.45%, so using alpha 1% with the same window cannot trigger any event via that rank rule. Even at 5% it produces a finite-step threshold. Governance of alpha and window length is therefore interdependent (Annex G).

C.3 Held-out 2017 fit, not a fresh hourly ASHRAE screen

The frozen 2016 models were evaluated out of time on **2017 daily aggregates** for 106 Fox buildings. The resulting median CV(RMSE) is **13.16%** and median NMBE is **−1.45%**. The interquartile CV(RMSE) range is 9.61%–18.02%, and 96.2% of buildings have a daily-aggregated CV(RMSE) no greater than 30% in the saved report. The daily aggregation matters: ASHRAE Guideline 14's familiar hourly thresholds cannot be treated as a direct pass/fail test on these daily values.

Evidence: src/e2_out_of_time.py, results/e2_out_of_time.json. This result strengthens the baseline discussion but does not eliminate model-selection bias or prove transferability to Dubai.

C.4 E3: persistent under-reporting versus physical relocation

Manipulations are injected into otherwise observed Fox metering traces from a random onset. One scenario multiplies the target building's later consumption by (1 − s); another replaces its post-onset signal with the real readings of a **different donor building**, scaled so the donor's 28-day pre-onset mean equals (1 − s) times the target's pre-onset mean. The latter is a model of a device-to-premises mismatch: the donor's own temporal drift is retained, so a nominal 20% target is not identical to uniformly shaving each future reading by 20%.

At a common test date 90 days after onset, the slow gate computes each building's residual shift against a frozen committed reference quarter and ranks it against 106 same-site peers. The one-sided rank targets **under-reporting** for green-credit claims; a two-sided alternative measures a different decision question. One building is manipulated per replicate, with 1,000 replicates per size and a fixed seed.

Injected magnitude	Under-report, one-sided	Relocation, one-sided	Under-report, two-sided
5.0%	9.9%	23.3%	3.5%
10.0%	18.9%	32.9%	2.2%
15.0%	37.5%	45.9%	5.5%
20.0%	66.6%	57.2%	9.4%
30.0%	88.6%	79.3%	43.4%
40.0%	97.2%	92.0%	84.6%

Evidence: src/e3a_quarterly.py, results/e3a_quarterly.json, and src/e4_review_budget.py.

Clean quarter-to-quarter residual drift has a logged standard deviation of 13.3%. Under-report detection is 66.6% at a 20% injection and 88.6% at 30%. A further 1,000 runs per size with a new seed measured the points in between (table below): detection reaches 80% at about a 25% under-report (80.7%), and the repeated 20% and 30% points (67.7%, 89.9%) agree with the first sweep within Monte Carlo error. This is a Fox-specific offline estimate, not a universal minimum detectable manipulation; population size, baseline error, seasonality, peer honesty, onset and test direction all matter.

Injected magnitude	Under-report, one-sided	Monte Carlo SE	Clean false-flag rate
20.0%	67.7%	±1.5 pts	4.1%
22.5%	75.7%	±1.4 pts	4.0%
25.0%	80.7%	±1.2 pts	4.0%
27.5%	85.0%	±1.1 pts	4.0%
30.0%	89.9%	±1.0 pts	3.9%

Evidence: src/e3a_extra.py and results/e3a_extra.json (seed 20261106; same one-sided quarterly test, 90-day horizon, alpha 5%, 106 buildings, 1,000 runs per size).

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C.5 E3: manipulation prevalence

A second experiment injects 20% under-reporting in an increasing share of the cohort. The peer reference is then progressively contaminated, and ability to identify a target declines.

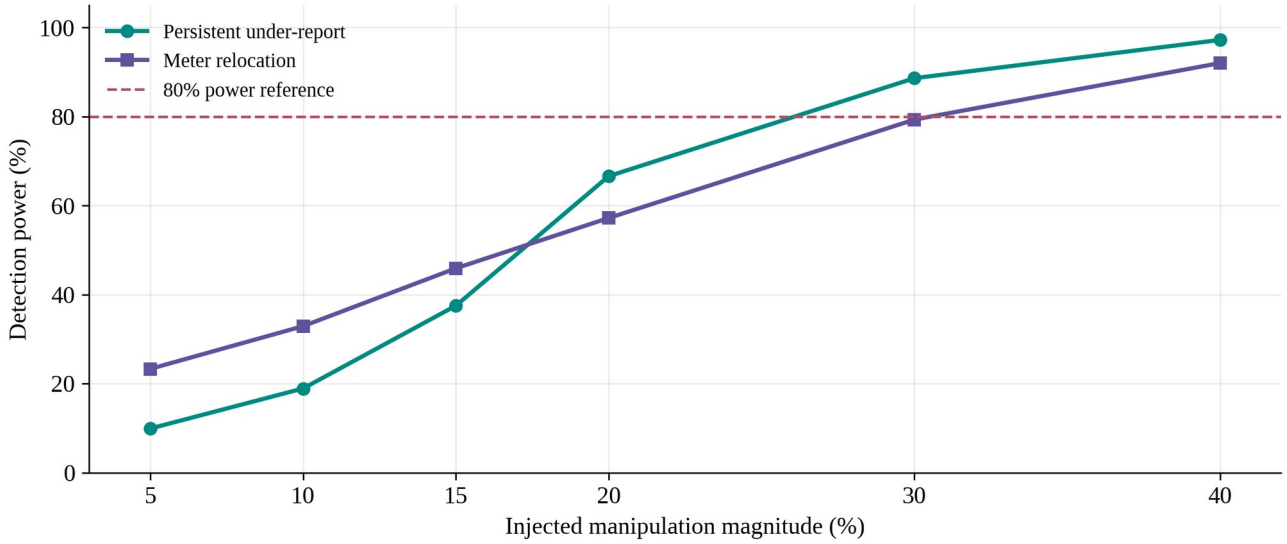


Fig. 2 Detection power as a function of injected manipulation. The experiment is an offline simulation, not field-detected fraud.

Manipulated share	Manipulated buildings	20% under-report power
1%	1	70.0%
5%	5	48.5%
10%	11	31.5%
20%	21	19.7%

Evidence: results/e3a_quarterly.json, key prevalence_sweep_shave20; 300 Monte Carlo repetitions per prevalence point. The one-manipulated-building row is 70.0% here versus 66.6% in the 1,000-run size sweep because Monte Carlo streams and repetition counts differ; those are distinct estimates, not an internal arithmetic contradiction.

C.6 E3: fast-gate contamination over time

For a sustained 20% under-report, the fast rolling window initially reacts: the saved curve records a **54.0%** flag rate on onset day. During days 1–6 the rate declines as manipulated readings enter the reference set; after approximately 28 days it returns toward its unlabelled clean background rate. This is why the fast gate cannot reliably protect a lender from sustained adaptive manipulation. The slow gate uses a precommitted reference and does not absorb the same drift; it nonetheless has limited power below the E3 signal threshold.

*Evidence: results/e3a_results.json (curves.shave_20.fast), src/e3a_gates.py. The fast-gate curve is an **onset-day watch signal probability**, not quarterly power and not a funded human-review case count.*

C.7 E3b: 3 low-cost manipulations

Challenge	Observed result	What the test actually supports
Exact / near copy	100% exact-match; 100% near-copy at 1% noise; 3.8% false flag	A simple exact rule and an offline similarity test may suffice
Optimistic rounding	98.1% of rounding detected; mean bias −1.94%; worst −4.29%	Detects digit pattern, not the magnitude of hidden bias
Selective deletion	Hottest month was worst only 4.7% (vs 8.3% chance)	Completeness rule needed; intent unprovable

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*Evidence: results/e3b_results.json, src/e3b_statistical.py. The result on selective deletion shows only that omitted adverse months cannot be reliably inferred from the remaining monthly pattern; it does **not** imply omission is harmless. A precommitted mandatory-input register, completeness rule and negative membership proof are necessary independent controls.*

C.8 E4: what detection costs in reviews

The slow quarterly peer-rank rule selects alpha × n cases per test, rounded down, by construction (assuming distinct ranks). For the Fox cohort of 106 buildings, at alpha 5% it selects 5 records on each of 3 observed 2017 test dates: 30 June, 30 September and 31 December. Annualising the *selection rule* to a hypothetical 4-quarter operating year gives 4 × (5/106) = 0.1887 selected cases per building-year. This is a budget allocation implied by the decision rule, not an empirically measured false-positive rate, observed staffing workload or count of fraudulent buildings.

Alpha	All flagged reviews / bldg-year	Honest flag estimates / year	Power: under-report 20%	Power: under-report 30%
1%	0.0377	0.0348	8.3%	26.4%
2%	0.0755	0.0632	33.8%	79.1%
5%	0.1887	0.1652	66.6%	88.6%
10%	0.3774	0.3512	78.1%	92.9%

Evidence: results/e4_results.json and src/e4_review_budget.py. The whitepaper and the workbook use 0.1887 rounded to 0.189.

A fast daily gate used as a human-case trigger was evaluated separately, with a 3-flags-in-7-days rule. Under the stated parameters it opens far more cases (below); these figures do not describe the operating 28-day watch-only configuration.

Window	Alpha	Cases / building-year	20% burst opened ≤ 7 d
28 d	5%	2.011	32.4%
56 d	5%	3.104	47.7%
56 d	2%	1.294	25.5%
112 d	5%	4.888	54.4%
112 d	2%	1.659	38.3%
112 d	1%	0.691	18.1%

Evidence: results/e4_results.json, key fast_gate_as_review. The 112-day, 1% configuration yields 0.691 cases per building-year and an 18.1% 7-day response to a 20% burst; it is not the 28-day, 5% watch-only case load.

C.9 Statistical validity and next tests

The saved outputs and source code support internal arithmetic consistency, seasonal sensitivity of observed alert rates, and offline detection-power versus case-budget trade-offs. On 6 November 2026, E2–E4 were re-run from the 3 raw BDG2 files (SHA-256 match) with the published scripts, and all 11 result files matched the published ones byte for byte (logs_2026-11-06/rerun_e2_e4/COMPARE.txt). The evidence does not demonstrate real-world false accusations on verified honest properties, prospective Dubai validity, or resilience when the committed reference quarter is manipulated before commitment. The next tests are labelled-onset evaluation, out-of-time site validation, cross-climate calibration, committed-baseline governance, small-peer-group analysis and 4 prospective quarter ends.

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ANNEX D

Cost Model, Assumptions, and Sensitivity Analysis

Submitted as part of the Project

DIYAR: A Provenance-Graded Evidence Layer for Sustainability Claims in the Built Environment

Fatemeh Zahra Bazrgar

Faculty of Humanities and Social Sciences, Kharazmi University

No. 43, Shahid Mofatteh Ave., Tehran, Iran

github.com/Akedia-tokenize/Diyar-IBCOL2026

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D.1 Scope and accounting convention

This annex documents the 6-tab cost workbook DIYAR_cost_model.xlsx. The reference scenario is 10,000 buildings in Dubai and a 1-year operating budget, with onboarding amortised over 10 years. Costs are in nominal USD and combine cited illustrative 2026 infrastructure prices, assumptions for staff, inspection and onboarding, and the E4 case rate. This is a modelled estimate, not invoices from a field deployment.

The workbook tabs are *Read me*, *Annual cost*, *Inputs*, *Custody deposit*, *Review vs alpha* and *Sensitivity*. The reference cells are Annual cost!B11 = $1,785,795.1231, Annual cost!C11 = $178.5795 and Inputs!B22 = 0.189 review cases per building-year, which is E4's 0.1887 rounded for entry.

D.2 Reference annual expenditure

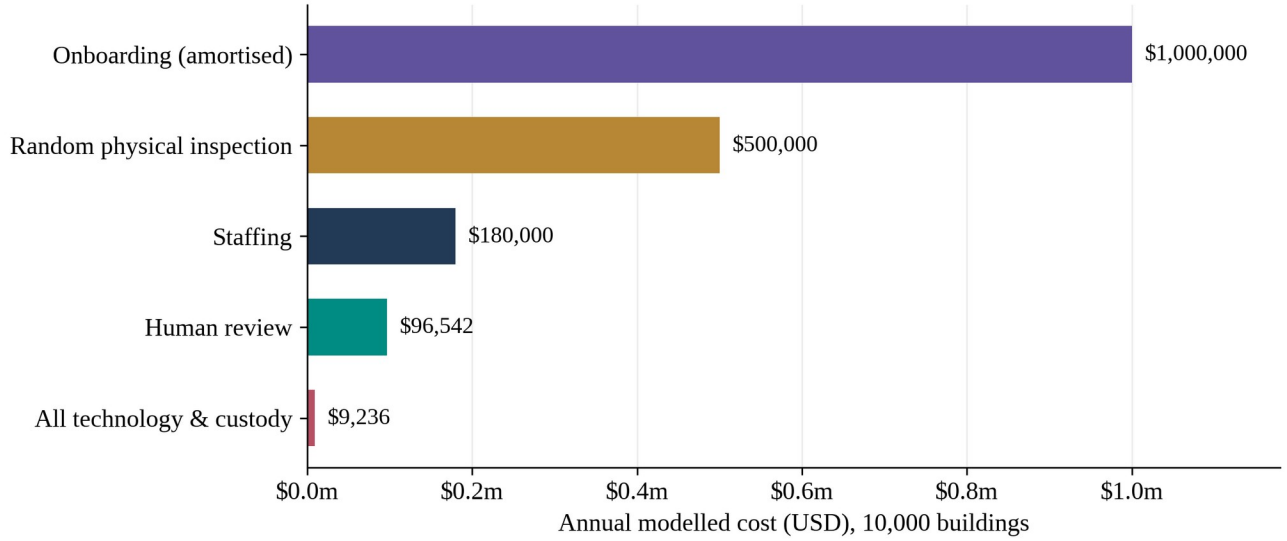


Fig. 1 Cost drivers for the illustrative 10,000-building scenario. Onboarding, inspection and staffing are assumptions, not measured operating expenses.

Annual line item	USD / year	Evidence class
Onboarding amortised	$1,000,000	Assumption; $1,000/building over 10 years
Random physical inspection	$500,000	Assumption; 5% × $1,000
Operator staffing	$180,000	Assumption; 3 FTE × $60,000
Human review	$96,542.40	E4 0.189 rounded × staff-rate assumptions
Permissioned validator VMs	$7,064.16	Quoted illustrative cloud rate, not deployed
Gate / replay compute	$1,766.04	Extrapolated hardware estimate
60-year custody deposit for one year of data	$393.21	Modelled storage prices and assumptions
Public-chain anchor	$29.31	Gas price and measured contract gas
Notary committee	$0	Scenario assumes no Dubai portal notary
TOTAL	$1,785,795.12	Workbook Annual cost!B11
Per building-year	$178.58	Workbook Annual cost!C11

*Evidence: DIYAR_cost_model.xlsx (Annual cost, Inputs) and results/e4_results.json. Technology costs in the workbook sum to **$9,252.72, around 0.518%** of the total; people, inspection and onboarding sum to **$1,776,542.40**, around 99.482%. These are category accounting shares under this scenario, not claimed future market shares.*

The 3 largest assumed categories are onboarding ($1.00m), random inspection ($0.50m) and staff ($0.18m): $1.68m, or about 94.1% of total. The review line combines an E4 selection frequency determined by a finite peer-rank quota, annualised from 3 tested quarter ends, with assumed reviewers/hours, sourced illustrative salary and productive-time assumptions. Its numerical precision is not evidence of observed review work or real field costs.

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The notary line is $0 because the Dubai scenario reads utility data directly and needs no A2− endpoint proofs. A path through a web portal, such as an EU utility portal, would run a committee of 3 notary nodes at the validator VM price: setting Inputs!B17 to 3 gives 3 × $147.17 × 12 = $5,298.12 a year, or $0.53 per building-year, and the total becomes $1,791,093.24, or $179.11 per building-year.

D.3 Human review calculation

At alpha 5%, results/e4_results.json gives 0.1887 total flagged cases/building-year, based on 3 observed quarter ends annualised to 4. The workbook enters **0.189**. Review labor is assumed at 2 independent reviewers, 1 productive hour each, and an effective **$25.54/hour** from a UAE energy engineer salary of AED 117,916/year, pegged AED/USD exchange rate 3.6725, employment loading 1.4×, and 1,760 **productive** hours/year. The workbook derives **$96,542.40/year** for 10,000 buildings under those specific choices.

The employment 1.4× factor must exclude paid leave if the 1,760-hour denominator already accounts for leave; otherwise the same cost is counted twice. This distinction is documented in the workbook but still warrants an external compensation-cost review. A case duration of 1 hour per reviewer is untested in practice. Claims of precise $/case field cost require measured reviewer times from the pilot.

D.4 Annualisation and alternative-alpha budgets

Alpha	Review cases / building-year	Case cost / building-year ($)	20% power	30% power
1%	0.0377	$1.93	8.3%	26.4%
2%	0.0755	$3.86	33.8%	79.1%
5%	0.1887	$9.64	66.6%	88.6%
10%	0.3774	$19.28	78.1%	92.9%

Evidence: results/e4_results.json. The table uses the exact 0.1887 values rather than the 0.189 rounded scenario input. The near-linear case budget is a consequence of finite same-site peer ranks and alpha. It does not prove that all flagged properties are fraudulent, nor that the review work can be completed at the assumed 1-hour rate. The separate fast-gate watch signals do not trigger desk review and are not counted here.

D.5 Inspection and onboarding sensitivity

The workbook assumes inspection of **5%** of buildings each year at $1,000/visit and amortised onboarding at **$1,000 per building over 10 years**. Varying the inspection share between 2% and 10% and visit price between $500 and $2,500, with other lines fixed, produces the following **per-building-year** outcomes:

Inspection share	Visit $500	Visit $1,000	Visit $2,500
2%	$138.58	$148.58	$178.58
5%	$153.58	$178.58	$253.58
10%	$178.58	$228.58	$378.58

Evidence: workbook Sensitivity tab. If a high-grade DEWA installation registry removed the onboarding line, the cost would move toward about $78.58 per building-year with everything else unchanged; this is a conditional upside, not a confirmed DEWA arrangement. Doubling onboarding raises the scenario toward $278.58. No blockchain throughput optimisation can offset an error in the inspection or onboarding estimate.

D.6 Anchor gas and crypto-price sensitivity

The anchoring figure uses the gas measured on the anchor contract, which keeps each day’s root in state and recomputes each entry: 58,929 gas for a typical daily anchor (deployment, once, 427,275 gas), one per day for the entire system, at ETH/USD $2,725 and gas price 0.5 gwei. The annual cost is $29.31, independent of the number of

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buildings in the batched root. The earlier 25,000-gas basis assumed an event-only anchor; storing roots in state costs about 2.36 times as much.

ETH/USD	0.5 gwei	5 gwei	20 gwei	50 gwei
1882	$20.24	$202.40	$809.60	$2,024.00
2725	$29.31	$293.06	$1,172.25	$2,930.61
8000	$86.04	$860.36	$3,441.45	$8,603.63
15000	$161.32	$1,613.18	$6,452.73	$16,131.81

Evidence: workbook Sensitivity tab and Inputs!B19:B21; gas from anchor/test.mjs in DIYAR-ref (17 tests against the reference ledger; anchor/gas.json). The gas is measured on a local EVM under Cancun rules: the contract is not deployed, so there is no public-network receipt yet. At the worst corner, ETH $15,000 and 50 gwei, anchoring costs $16,131.81 a year, or $1.61 per building-year. The verification and archival costs of each receipt/header proof bundle are not measured separately in this workbook.

D.7 Custody deposit for a 60-year horizon

For one new year of data from 10,000 buildings, the workbook gives a $393.21 illustrative deposit for the full 60-year storage lifecycle: 2 hot years, 58 archive years, integrity reads and a 20% reformat and migration overhead. The inputs specify a 1.4× RS(10,4) expansion, cloud hot storage at $0.023/GB-month and archive at $0.00099/GB-month. The deposit covers signed meter telemetry only (11.25 TB in one copy), not full building-evidence or BIM archives. It is paid at origination rather than depending on the operator remaining solvent; prepayment alone does not prove governance, key availability or future custody-service pricing for 60 years.

D.8 Evidence classes and unpriced risks

- **Measured/prototype-derived:** E4 review-case frequency, measured record/signature sizes, prototype CPU context.
- **Illustrative external prices:** validator VM, energy-engineer salary, cloud storage, gas and ETH scenario values; independently refresh before commercial use.
- **Assumptions:** onboarding, inspection frequency and unit cost, team of 3, review labour, migration and retention.
- **Not priced or validated:** sovereign and legal data fees, approved trust-service credentials, issuer and installer key registries with revocation governance and installer validation, physical meter installation at scale, litigation, consortium governance, security operations, key-recovery drills and 60-year capital-risk costs.

The whitepaper’s figure of about $179 per building-year is the workbook’s Annual cost!C11; it is a conditional planning estimate, not an observed unit price.

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ANNEX D / D.9 | PILOT-TO-SCALE COST AND FUNDING REQUIREMENTS

Integrated supplement to the 10 November 2026 DIYAR whitepaper v2.7 and original Annex D. The original model, E1-E4 outcomes and 112-value reconciliation are preserved. All monetary figures are USD.

D.9.1 What the original USD 178.58 measures

The existing six-tab cost model reports $1,785,795.12 per year for 10,000 buildings ($178.58/building-year), under a full operating-stack scenario with three staff FTE, four Besu validators, one compute VM and 0 notary nodes. It is a normalized annual economic cost, not the cash required to onboard a complete portfolio in its first year.

Its assumed $1,000 one-time onboarding cost per building is annualized over ten years ($100 per building-year). When all buildings enter in Year 1, the modeled cash requirement must replace that $100 fraction with the full $1,000. Therefore, for N newly onboarded buildings: Year-1 priced cash subtotal = normalized annual economic cost + $900 × N. This is an accounting bridge, not an externally validated new price.

D.9.2 Mechanical scale comparison

Same full stack	Annual economic cost	Year-1 priced cash subtotal	Year-1 cash per building
100 buildings	$204,829	$294,829	$2,948
1,000 buildings	$348,553	$1,248,553	$1,249
10,000 buildings	$1,785,795	$10,785,795	$1,079

The 100- and 1,000-building rows retain the full reference configuration and are deliberately not optimized pilot budgets. The 10,000-building annual amount agrees with the unchanged source cost workbook to the cent; values in the table are rounded.

D.9.3 Lean pilot illustration, not a commitment

100-building scenario	Original full stack	Lean phase-1 hypothesis
Staff FTE	3	1 (not workload-validated)
Besu validators	4	0 (Besu deferred)
Compute / notary nodes	1 / 0	1 / 0
Annual economic cost	$204,829	$77,765
Year-1 priced cash subtotal	$294,829	$167,765

A phase-1 witnessed transparency log and public-chain anchoring design is consistent with Annex F, which defers Besu. However witness cost is not included, and one FTE has not been validated as sufficient. The USD 167,765 lean figure is only a partial priced cash subtotal, not a complete funded pilot budget.

D.9.4 Human operations at scale

At 5% random annual physical inspections and 0.189 quarterly review cases per building-year, 100 buildings imply approximately 5 inspections, 18.9 review cases and 100 initial onboarding events if the cohort joins in Year 1. The 10,000-building scale implies 500 inspections, 1,890 cases and 10,000 onboarding events. Inspection capacity, contractor availability and consent lead times are not established by the current FTE assumptions.

ANNEX D / D.9 | FUNDING, PROCUREMENT AND ASSURANCE

The finance test for implementation is not just average annual cost. It is which cash obligations arise in the first year, who funds them, and what evidence makes the pilot financeable.

D.9.5 Unpriced prerequisites

Dependency	Evidence required before a defensible funding request
Data rights and access	Named utility/operator integration path, building permissions, consent, API entitlement and access SLA.
Integration and deployment	Scoped quote for source ingestion, registry mapping, error handling, hosting and acceptance testing.
Physical binding and inspections	Qualified installation-verification quotation; reconcile scope against the assumed $1,000 onboarding and $1,000 inspection charges without double-counting.
Witnesses, HSM and keys	Independent witness operations, PKI/HSM procurement, issuance, recovery, rotation and breach response.
Procurement and law	Contracting, privacy, insurance, liability, supervision and explicitly priced contingency.

The two large field-cost assumptions ($1,000 onboarding and $1,000 inspection) are not verified Dubai meter-binding supplier quotes. A generic energy-audit or building-inspection advertisement has materially different scope and cannot silently validate them.

D.9.6 Funding and verifier independence

Potential pilot sponsors include an asset owner, lender, valuer, or public innovation program. The source model does not establish a signed buyer or sponsor. Any verifier fee should be fixed or based on work performed, and contractually independent of whether an evidence claim passes, fails or receives a lower grade. Outcome-linked fees could undermine neutrality.

D.9.7 Four financing decision gates

Gate	Deliverable	Decision
1. Eligibility	100-building cohort, consent and metering access	Confirm scope
2. Delivery capacity	Staffing, witness model, inspection and integration plan	Approve design
3. Procurement	Third-party RFQs for integration, fieldwork, HSM, legal and contingency	Approve cash envelope
4. Assurance	Independent replay and fail-closed test evidence, milestones and acceptance criteria	Authorize pilot

D.9.8 Evidence basis and boundary

The figures above derive only from DIYAR_cost_model.xlsx (Inputs, Annual cost, Custody deposit and Review vs alpha) and the separate formula-driven pilot-to-scale financial supplement. They do not modify the original six-sheet model, the USD 178.58 baseline or its E1-E4 scientific claims. They do not prove a Dubai utility deployment, site access, regulatory approval, staffed operational sufficiency, or mainnet anchoring. The costs omitted above must be quoted before a full cash funding request can be represented as complete.

Available supporting package: https://github.com/Akedia-tokenize/Diyar-IBCOL2026 · original workbook under Reference & Tests · scenario register under Results / pilot-to-scale. The original whitepaper's Section V financial claims are unchanged; this annex provides their pilot-stage accounting qualification.

ANNEX E

Long-Horizon Cryptography, Storage, and Custody

Submitted as part of the Project

DIYAR: A Provenance-Graded Evidence Layer for Sustainability Claims in the Built Environment

Fatemeh Zahra Bazrgar

Faculty of Humanities and Social Sciences, Kharazmi University

No. 43, Shahid Mofatteh Ave., Tehran, Iran

github.com/Akedia-tokenize/Diyar-IBCOL2026

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E.1 Time horizon, measurement basis and signature strategies

This annex provides the engineering assumptions behind Section IV (cryptography) and Section V (custody) of the paper. The storage horizon is **60 years** for **10,000 buildings** with 5 channels per building and **96 readings per channel-day**. The canonical reading record is 43 bytes; the raw ECDSA P-256 signature is 64 bytes (r‖s). These sizes were exercised in E1. However, the source BDG2 dataset is **hourly**, not 15-minute: the quarter-hour 60-year forecast is a design extrapolation, not a measurement from 10,000 real 15-minute meters.

The E1 channel census records a mean of **1.852 channels** among 1,631 source buildings. The 5-channel case is a deliberately conservative scenario. A measured 1-year, 5-channel example used Bobcat_education_Dylan and 42,708 hourly readings (missing hours versus a complete year are possible). Signature byte counts are measured by the signing test; the 60-year data volumes are formula-derived.

E.2 Alternative 60-year signing designs

Strategy	Description	60 years, one copy (TB)
S0	Raw canonical records	4.520
S1	ECDSA each reading	11.248
S2	SLH-DSA-128s per reading	837.071
S3	SLH-DSA-256s per building-day	17.772
S4	ECDSA reading + one global SLH daily root	11.248
S5	One ECDSA per channel-day, conditional on DLMS	4.591

*Evidence: results/e1_summary.json, results/e1_storage.csv, src/e1_storage.py. Values are decimal terabytes and use 365 days/year. S4 uses one **system-wide** daily SLH-DSA-256s signature, not a per-building root signature. The storage difference between S1 and S4 is just **21,900 signatures × 29,792 bytes ≈ 0.652 GB** over 60 years. The total S4 footprint is **11.2485 TB**, reported as **11.25 TB**; 3× replication gives **33.75 TB** and RS(10,4) gives **15.75 TB** when displayed to two decimal places.*

S5's 4.59-TB figure presumes a meter/utility protocol that can aggregate readings and provide a legally meaningful device signature per channel-day. The repository explicitly states that this must be checked against the actual DLMS/COSEM security specification, so it is an **unapproved optimisation**, not the operating model. S2 at **837.1 TB** shows why signing every individual reading with a post-quantum primitive is disproportionately expensive.

E.3 Exact signatures, commitments and verification

For each A1 prototype reading, the device signature is an ECDSA P-256 signature over an implementation-defined canonical representation including device/field, time and value. The original energy data did not carry those signatures; the reference run adds them with simulated keys. A daily sparse-Merkle state root binds the field records. The tree uses 256-bit registered-field keys and domain-separated leaf, internal-node and empty-state hashes. Membership and non-membership proofs allow selectively revealing one **exact** committed field without disclosing every other field.

The daily ledger signs a chained state root with ECDSA P-256 and SLH-DSA-256s, parameter sha2_256s. The E1 measurement reports a **29,792-byte** SLH-DSA-256s signature and a **7,856-byte** SLH-DSA-128s signature, each with successful internal verification. In the reference run all 91 daily roots carry both signatures and verify on replay with the slh-dsa library; the 91 signatures of the archived earlier run also verify with OpenSSL 3.5.5. This provides a post-quantum signature on the commitment under the stated hash and signing assumptions. It does **not** make the physical sensor, the operator key custody, AES, or a future chain history automatically quantum-safe.

The commitments rely on 2 properties of SHA-256, and quantum computers affect them differently. Grover's algorithm lowers preimage and second-preimage search from about 2^{256} to about 2^{128} work. Collision search already

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costs about 2^{128} classically (the birthday bound). The best-known quantum collision algorithm, by Brassard, Høyer and Tapp, needs about 2^{85} queries but also about 2^{85} entries of quantum-accessible memory; counted as hardware × time it is less cost-effective than classical parallel collision search (Bernstein, 2009). NIST's post-quantum security categories rank collision search on a 256-bit hash (category 2) between AES-128 and AES-192 key search (categories 1 and 3). Both properties therefore keep about 128-bit security against known quantum attacks, which is the whitepaper's conclusion. SLH-DSA's own security rests on the difficulty of finding preimages for its hash functions and on related properties. The implementation's canonical JSON is locally deterministic but requires an interoperable canonicalisation standard and cross-language vectors before relying on 60-year third-party replay.

Sources: G. Brassard, P. Høyer and A. Tapp, "Quantum cryptanalysis of hash and claw-free functions," LATIN '98, LNCS 1380, 1998; D. J. Bernstein, "Cost analysis of hash collisions: Will quantum computers make SHARCS obsolete?," SHARCS 2009; NIST, Post-Quantum Cryptography Standardization, evaluation criteria (security strength categories); IETF draft-ietf-lamps-x509-slhdsa, Section 4.

E.4 RS(10,4) versus triple replication

The proposed Reed–Solomon scheme uses 10 data shards plus 4 parity shards and expands encoded storage by 1.4×. RS coding replaces, rather than supplements, 3 complete replicas in the top-line comparison. An illustrative allocation across 4 independent custodians is 4 + 4 + 3 + 3 shards. Because any 10 of the 14 suffice, losing or withholding the largest single custodian leaves 10 shards. Losing 2 custodians could leave only 6 to 8; this is not a 2-custodian tolerance claim.

Tolerating the loss of 2 custodians requires at most 2 shards per custodian, hence at least 7 custodians. This is a design illustration; the general recovery condition is "at least 10 recoverable shards under the specified outage set". Local durability and erasure coding across independent legal entities defend against different failure modes and are not additive capacity multipliers.

Evidence: results/e1_storage.csv. The topology, the distributed encrypt-then-encode workflow and multi-custodian recovery are not implemented in DIYAR-ref.

E.5 Encryption and the 60-year key problem

Systematic RS coding does not guarantee confidentiality: 10 of the 14 shards retain information-bearing data unless it is encrypted first. The proposal therefore encrypts before erasure coding (with AES-256 as the candidate symmetric primitive), then distributes ciphertext shards. The residual trust becomes **key availability and key control**. If the key is lost, a perfectly intact 15.75-TB archive may be unusable; if a single operator controls the key, custody dependency returns as key monopoly.

The design uses one data-encryption key **per building-year**, with threshold Shamir shares among custodians. In the 10,000-building/60-year model, this implies **600,000** separately governed key lifecycles. The point of this granularity is blast-radius limitation when one slice is reconstructed for a lawful replay. Basic threshold sharing does not prevent the entity that recombines a key from retaining a copy; it does not provide end-to-end threshold decryption by itself. Nor does Shamir sharing guarantee secure key-share distribution or recovery under malicious-custodian behaviour.

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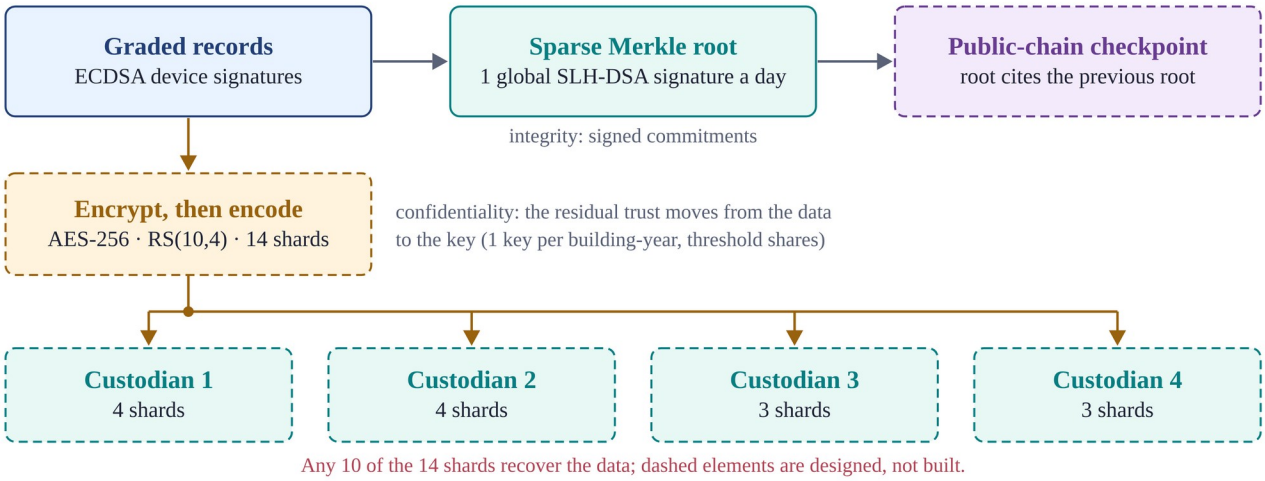


Fig. 1 Separation of signed commitments from long-horizon encrypted custody. 4-custodian RS(10,4) distribution is a proposed deployment, not an executed prototype test.

E.6 Share transport and long-term confidentiality

A recorded key-share transmission protected only by classical ephemeral elliptic-curve key exchange can be vulnerable to future “harvest now, decrypt later” capability. The proposed transport is a **hybrid ML-KEM + X25519** channel for transactions carrying shares and any decrypted fields. This separates long-lived confidentiality from integrity, which only needs to survive immediate verification. The alternate design uses pre-positioned symmetric secrets exchanged out of band, at higher operational cost.

Shamir’s information-theoretic threshold property is not a complete post-quantum security proof for the deployed protocol: encryption, authentication, hybrid key exchange, recovery, revocation, member rotation, offline backup and the legal right to release shares also need audited mechanisms. Proactive re-sharing is necessary as operators change over decades; it is not implemented in the reference repository.

E.7 Public anchoring and retained verification bundle

The anchor contract (anchor/DiyarAnchor.sol in DIYAR-ref) stores each day’s root in state and requires every entry to cite the previous one. On a local EVM a competing entry, a repeated day and a second writer all revert (Annex B); the contract is not deployed on a public network. The design distinguishes this state-storing contract from a low-gas event-only transaction: the latter makes conflicting histories easier to detect but does not enforce continuity in the same manner. Its measured gas, 58,929 per daily anchor, is priced in Annex D. Signatures alone cannot prevent equivocation; that takes an external shared checkpoint or independent witnesses.

Because Ethereum execution clients may prune historical data, each anchor must be accompanied by transaction bytes, receipt, receipt inclusion proof, block header and a verification path to a recognised chain state. The bundle is estimated at a few KB per day; its size has not been measured. Availability after operator failure depends on independent witness/custody storage and long-term access to necessary verification headers, not just having written a hash once.

E.8 Platform-disappearance and operational acceptance tests

- Can an independent verifier obtain 10 shards and reconstruct a building-year without the original operator?
- Can the correct key-shares be lawfully recovered after one custodian disappears, and is recovery logged without retroactively changing original evidence?

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- Do signed roots and retained transaction/receipt proofs verify from an independently trusted checkpoint, even if the chain node has pruned bodies?
- Can records and model/calibration versions be decoded by a third-party implementation decades later?
- Can consent withdrawal, key rotation, retention mandates and legal holds be reconciled with immutable integrity commitments?

Only the sparse-Merkle proofs, root signing and bitemporal replay have prototype evidence, and the anchor contract has local EVM tests. RS custody, independent keys, legal-grade signatures and a deployed Ethereum anchor are deployment workstreams; the whitepaper labels them designed or not deployed.

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ANNEX F

Dubai Pilot, Data Access, Legal Conditions, and Milestones

Submitted as part of the Project

DIYAR: A Provenance-Graded Evidence Layer for Sustainability Claims in the Built Environment

Fatemeh Zahra Bazrgar

Faculty of Humanities and Social Sciences, Kharazmi University

No. 43, Shahid Mofatteh Ave., Tehran, Iran

github.com/Akedia-tokenize/Diyar-IBCOL2026

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F.1 Pilot premise and why Dubai is not yet validated

This annex expands the pilot and legal dossier behind Sections II, V and VI of the whitepaper. Its statements rest on the cited instruments, not on an independent legal opinion. The Dubai pilot remains a **conditional proposal**. Fox, the hot-site benchmark used in E2/E3/E4, is cooler and drier than Dubai; the pilot's first task is to determine whether the same baseline and calibration behaviour holds on authorised Dubai interval data. Baselines trained on signed Dubai records would also lift their grade: the 2016 BDG2 training data are unsigned CSV files, so the baselines and the performance figures built on them are graded C even though their hashes reproduce.

F.2 Regulatory and evidentiary map

Instrument / source	Claim in project dossier	Caveat
RICS ESG & Sustainability in Valuation (2025)	Effective 30 Apr 2026; sustainability factors considered in valuation	Evidence assessment is not an automatically accepted tokenised score
EU Taxonomy / EPBD	EPC alternatives include top-15% benchmark evidence; EPBD extends certificates	Country-specific applicability and datasets require check
EU 2024/825	Environmental claims compliance regime with 2026 transposition/applicability milestones	Do not conflate transposition and application dates
UAE Federal Decree-Law 11/2024	Climate reporting obligations and record retention	Scope and verification guidance need local counsel
Dubai Law 3/2026	Quality/safety certifications and municipality database	Not an energy-performance certification
UAE Evidence Law 35/2022	Electronic evidence may be admissible subject to conditions	A Merkle proof is not judicial acceptance by itself
UAE Electronic Transactions Law 46/2021	Qualified/approved electronic signatures subject to approved providers	Simulated keys have no approval status

Evidence: whitepaper references [5]–[8] and [23]–[25]; UAE Federal Decree-Law Nos. 11/2024, 35/2022 and 46/2021; Directive (EU) 2024/825. The dossier distinguishes Dubai's building quality and safety law from an energy-performance rule, and RICS evidence duties from an assumption that an automated sustainability valuation factor may be inserted without professional judgment.

The whitepaper cites the RICS sustainability valuation standard taking effect on 30 April 2026, EU Taxonomy's "adequate evidence" approach to top-15% building performance in relevant circumstances, UAE reporting obligations, and Dubai Law No. (3) of 2026 concerning structural quality and safety. Each is used as a **use-case motivation**, not evidence that a regulator has endorsed DIYAR. Applicability to a specific building class, data subject and local evidence law must be independently checked before a real pilot.

F.3 Practical access to authoritative data

DEWA has a large automated meter base, and its R&D Centre has used interval data in published work with consenting subjects (Manandhar et al., *Energies* 16(1), art. 285, 2023; whitepaper reference [23]). This is precedent for research use under its own arrangements, not permission for DIYAR or evidence of a public third-party API.

We found no general, publicly documented third-party API that grants full building-level interval access in Dubai; the exact permissions remain an open dependency. Retail customer-facing Smart Living views and aggregate public datasets do not resolve this. The intended route is a formal data-sharing agreement, lawful consent (owner and potentially tenant), minimum necessary fields, retention terms, and a reproducible anonymisation/de-identification and matching protocol reviewed with the data controller.

A further central dependency is whether an authoritative DEWA installation or meter-to-premises registry can supply a high-grade device-location binding. This could reduce both physical misbinding risk and the largest onboarding cost assumption. Until it is verified, installation binding stays at B. DIYAR-ref's signed issuer claims and installation certificates are enforceable software checks with simulated, pinned keys; they establish no DEWA

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API entitlement, legally recognised issuer identity, utility-signed 15-minute record, metering topology or lawful basis. These remain pilot go/no-go dependencies, and the utility's involvement does not by itself establish A1 device signatures or third-party access.

F.4 Data categories and attainable provenance grades

Data category	Plausible path	Key condition / residual trust
Electricity / water	A1 device or A2 utility	Meter signatures and certified property mapping are different questions
Tenancy / occupancy	A2 registry if authorised	Third-party Ejari access and consent not established
Service charges	A2 if Mollak access exists	Third-party entitlement not established
Green-building certificate	B / A2 feed	Issuer registry, currency and evidence scope
Embodied carbon	B at best in current proposal	Methodology, assessor and material provenance
Occupant behaviour	C in present design	Self-report limitations not removed by signing

The above grades are conditional targets, not verified production coverage. Specifically, A2 for tenancy and service-charge feeds depends on actual authorised access to **Ejari** and **Mollak** records; a registry's existence does not grant a third party rights to its contents. The project also refers to Dubai's **Al Sa'fat** green-building framework as a potentially useful certificate source. It is a certificate source, not continuous metered performance or a retroactive validation of a previously self-declared figure.

F.5 Legal admission is not automatic

Two instruments frame admissibility: the UAE Evidence Law (Federal Decree-Law 35/2022, Articles 53–55) and the Electronic Transactions and Trust Services Law (Federal Decree-Law 46/2021, including provisions on qualified and approved signatures). Even where statutory electronic evidence can have legal effect, a self-generated prototype ECDSA or SLH-DSA key is **not automatically an approved trust-service credential**. Legal admissibility may depend on source, integrity, certification, signatures, chain of custody and judicial or supervisory judgment. SLH-DSA in the current design provides an additional long-horizon integrity primitive, not a claimed special legal status.

The pilot requires local legal advice confirming controller/processor roles, lawful access, source consent, authorised signers, evidence preservation, dispute procedure, key custody, cross-border handling and the regulator's specific reporting rules. Court acceptance of a Merkle proof, or the deployment of Besu, is not presumed.

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F.6 Pilot phases with measurable exit gates

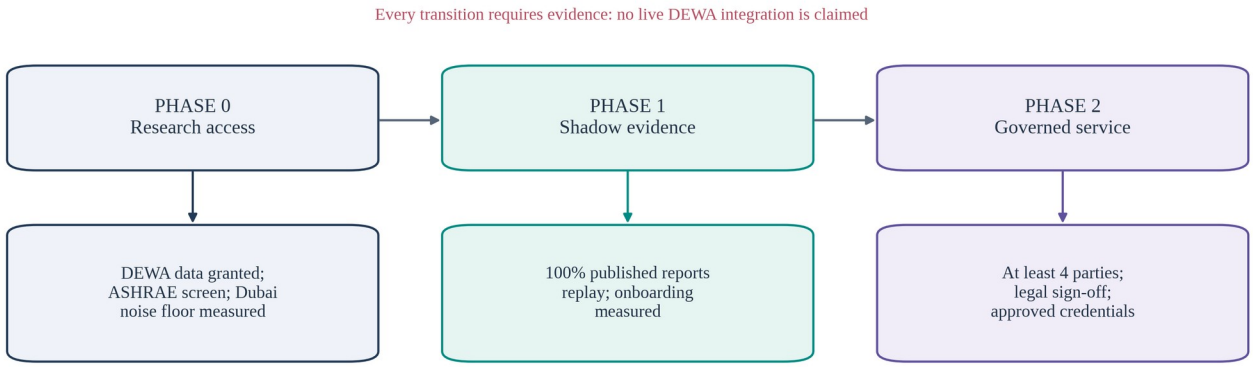


Fig. 1 Proposed Dubai pilot with explicit pass/fail gates and external dependencies. No utility deployment or legal approval is represented as complete.

Phase	Action	Evidence-based pass/fail gate
0: research data	Seek anonymous 15-min data under DEWA research collaboration; re-run E2/E3	ASHRAE 14 screening; each month rolling flag rate <10%; Dubai noise floor measured
1: shadow pilot	Opt-in buildings; attach evidence to reports without changing credit/valuation	Replay matches 100% of published reports; onboarding costs measured; registry binding tested
2: governed service	Independent consortium, approved signing, Besu only if justified	At least 4 independent parties; written counsel opinion; verified access/custody

*Evidence: whitepaper Section V (Pilot path). Phase 0 is primarily research replication; Phase 1 is a shadow process with **no automated lending or certification consequence**; Phase 2 is a conditional governed service. Gate failure should stop scale-up rather than be concealed by adjusting the dashboard.*

F.7 Open institutional questions

- **Trust-service approval:** Which authority approves qualified/approved trust-service providers and credentials for the proposed deployment, and what evidence format is accepted?
- **Utility meter security:** Which DLMS suite and actual export signing mode does DEWA employ? Signed raw values, authenticated feeds and building-location registry may carry different grades.
- **Tenancy and service charge:** Can a consenting third party receive machine-verifiable Ejari and Mollak records, and on what lawful basis?
- **Dubai Law 3/2026:** What is the authoritative gazette publication/effective date and which building classes follow which transition rules? No grace period is assumed until one is verified.
- **Installation registry and voltage topology:** Will DEWA provide a signed meter-installation mapping and, where relevant, neighbour/transformer voltage or energy-balance data to constrain physical relocation?

None of these dependencies is an existing partnership or a confirmed production interface.

F.8 Consent, output and dispute roles

The data owner may authorise a **specific purpose and set of evidence fields**; selective disclosure limits exposure of unrelated leases and tenant data. Public anchoring stores commitments, not the raw tenancy file. The product is an **evidence file** with provenance composition and replay receipts. The valuer retains independent judgment and signs a valuation; the lender/insurer makes its own regulated decision; a regulator may inspect traceability but is not a

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presumed consortium member. Under a disputed result, the grade and evidence register must permit review without allowing the original operator to rewrite prior state.

A binding gate for pilots is **operator disappearance**: the project should be able to reproduce a report using independent data, public receipts, recoverable keys and a published verifier after the operator exits. At present this is only partly demonstrated locally, and its organisational/legal components remain designed.

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ANNEX G

The Price of Alpha, Calibration Policy, and Fairness

Submitted as part of the Project

DIYAR: A Provenance-Graded Evidence Layer for Sustainability Claims in the Built Environment

Fatemeh Zahra Bazrgar

Faculty of Humanities and Social Sciences, Kharazmi University

No. 43, Shahid Mofatteh Ave., Tehran, Iran

github.com/Akedia-tokenize/Diyar-IBCOL2026

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G.1 Alpha is a resource and fairness parameter

The level alpha controls how aggressively the empirical peer-rank procedure allocates cases for review. It trades review capacity and the risk of interrupting owners whose wrongdoing has not been established against sensitivity to injected under-reporting. DIYAR therefore treats alpha, peer-set construction, calibration horizon and rule-effective dates as published governance parameters rather than hidden tuning defaults. An unlabelled rank flag does not establish fraud or a wrongful accusation.

G.2 Quarterly peer-rank gate: case load and detection

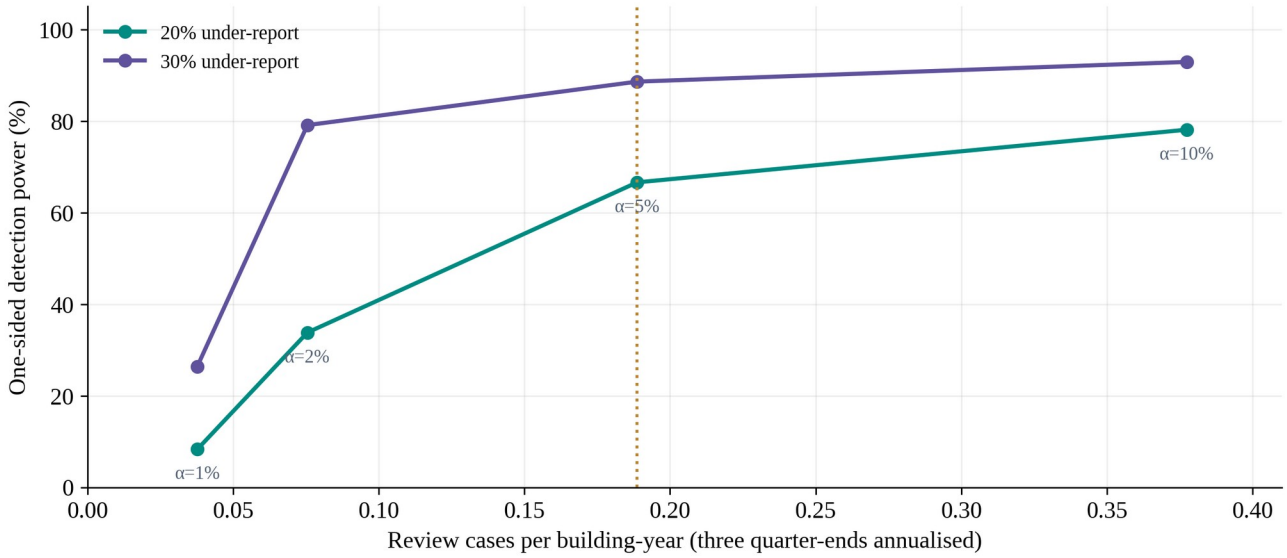


Fig. 1 Rank-determined case-selection budget versus injected-under-report detection power. The 0.1887 at α = 5% annualises the 5 cases selected at each of 3 tested quarter ends to a hypothetical 4-quarter year.

Alpha	Review cases / building-year	Case cost / building-year ($)	20% power	30% power
1%	0.0377	$1.93	8.3%	26.4%
2%	0.0755	$3.86	33.8%	79.1%
5%	0.1887	$9.64	66.6%	88.6%
10%	0.3774	$19.28	78.1%	92.9%

*Evidence: results/e4_results.json (slow_quarterly_gate, caseload_detail), src/e4_review_budget.py. Costs shown here use 2 reviewers, 1 productive hour each and $25.54 per hour. The workbook uses a rounded **0.189** reviews/building-year and therefore reports about $9.65 of reviews/building-year at alpha 5%; this annex shows precise E4 frequencies.*

The peer-rank procedure flags alpha × n peers per quarterly test, rounded down. For Fox n = 106, the alpha-5% rank produces **5 cases per test**, or 5/106 = 0.04717 cases per building each quarter. The annualised **0.1887** is obtained by multiplying by 4. The dataset contains 3 observed quarter-end test dates (end-June, end-September, end-December 2017), not an observed fourth quarter. Distinguish all flagged cases (0.1887) from estimated honest-only flags (0.1652); the latter is smaller in simulated manipulation runs because it excludes deliberately injected buildings.

Between the tested sizes, a further 1,000 runs per size at alpha 5% measured 75.7% detection at a 22.5% under-report, 80.7% at 25% and 85.0% at 27.5% (Annex C.4): the alpha-5% quota reaches 80% detection at about a 25% under-report.

The peer-rank procedure mechanically allocates cases, irrespective of whether the evidence is honest or fraudulent. Predictable selection volume is useful for budget planning but is not measured false-positive performance. The pilot

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therefore requires explicit escalation, appeals, quality-of-evidence checks and peer-group-size rules before any adverse decision.

G.3 Fast watch signals versus slow review cases

The running fast gate uses an alpha-5%, prior-28-day conformal score and emits **watch signals only**. Under nominal 5% behaviour, that is approximately 18 watch signals per building-year; they are not equivalent to 18 human reviews. E4 evaluated several **alternative** fast-gate case-opening configurations with a 3-flags-in-7-days escalation, and showed much higher personnel cost per unit of reliable detection. Such configurations are experiments, not the selected operating rule.

The slow gate is tested quarterly against a committed reference. It does not first require a fast warning: if a sustained 20% downward shift becomes part of the 28-day rolling calibration, the daily warning probability can return toward normal while the slow reference still detects a fraction of cases. Conversely, a brief interruption may trigger the daily rule/watch path but not a quarterly persistent-shift review.

G.4 Discrete p-values couple alpha and window length

With W exchangeable calibration scores and the empirical rank convention used here, the smallest achievable p-value is 1/(W+1). At W = 28, the minimum is **3.45%**. An alpha 1% cutoff therefore cannot flag a point on this 28-point empirical test; alpha 1% needs at least **99** calibration observations to make a p-value at or below 1% possible. Changing alpha without changing reference-window design can make the detector mathematically inert.

The main paper's alpha 1% versus alpha 5% detection-power comparison belongs to the **quarterly peer-rank** experiment, not a claim that a 28-day fast gate operates at 1%. A joint public policy should record window, minimum observations, cohort membership rules, alpha, versions and effective dates before evidence periods are assessed.

G.5 Seasonality, Mondrian and ACI

E2 documents 4 calibration designs on unlabelled real data. Same-season Mondrian calibration produced an overall **14.47%** observed Fox flag rate against nominal 5%, with a worst-month rate of 21.07%; it did **not** solve temporal calibration error merely because calendar season was matched. Rolling 28 days produced **5.33%**, worst month 8.36%; adaptive conformal inference produced **5.07%**, worst month 7.62%.

The engineering reading is that **recency and prediction-horizon alignment** were more important than same-season grouping for this dataset. ACI is designed to manage long-run alert frequency under adaptation; observing about 5% alerts does not independently label any record as false-positive or guarantee exact conditional coverage in every building. Mondrian grouping remains a candidate **fairness diagnostic**, not a demonstrated fix for seasonal validity.

Evidence: results/e2_summary.json and src/e2_conformal.py; method: Gibbs and Candès (2021), whitepaper reference [20].

G.6 Small groups and unequal error burden

A fairness policy should ask whether a chosen cohort overflags certain building types, regions, sizes or tenant profiles. But splitting data into smaller conditional groups reduces effective calibration sample size. A subgroup of m observations has minimum rank p-value 1/(m+1); if m is small, the requested alpha may not be attainable. The project should disclose (a) cohort definition and size, (b) effective threshold granularity, (c) monthly and building-level alert distributions, (d) whether grouped coverage was actually measured with verified labels, and (e) review appeals and protected-feature constraints.

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The analysis reports the distribution of **observed flag rates**, not verified wrongful accusations by protected group. Claims of equalised fairness should therefore remain a proposed governance objective. The smallest groups may require wider prediction sets or a longer calibration horizon; these are observable trade-offs, not errors to conceal.

G.7 Proposed parameter-change procedure (design, not deployed)

A governance vote publishes an alpha or peer-set update with model/configuration hash and a future effective date, then commits that change before the affected assessment period. Reviewers can reconstruct why a case was opened under the rule that applied at the time. The timelock is specified but not implemented in the reference software, and the anchor contract that would carry the commitment is tested only on a local EVM, not deployed; the procedure is an operational requirement for a later pilot, not a current on-chain feature.

The policy also reserves independent, score-agnostic random physical inspection as the only proposed path for catching **sub-threshold manipulation** that statistical peer tests cannot consistently detect. That inspection rate itself belongs in the public cost model and merits a separate equity assessment.

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ANNEX H

Prior Art and Threat-Model Comparison

Submitted as part of the Project

DIYAR: A Provenance-Graded Evidence Layer for Sustainability Claims in the Built Environment

Fatemeh Zahra Bazrgar

Faculty of Humanities and Social Sciences, Kharazmi University

No. 43, Shahid Mofatteh Ave., Tehran, Iran

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H.1 Framing: the nearest real alternatives

DIYAR does **not** invent transparency logs, sparse Merkle trees, hash-based signatures, conformal prediction, key transparency or endpoint notarisation. Its claim is the integration of field-level provenance grades, residual trust disclosure, valid-versus-record time, committed derivation/replay, and an independently checkable history of evolving building evidence. The relevant baseline is not a naive spreadsheet; it is a **well-designed signed database with independent witnesses, a verifiable map, version history and a replay engine**. Any incremental public-chain value must be evaluated against that realistic baseline.

A claim that “a transparency log is a list whereas DIYAR is a map” would be too broad: key transparency already operates on verifiable maps (CONIKS and later designs). DIYAR is distinguished by building-evidence semantics and bitemporal derivation, not by a new data structure.

H.2 Feature and trust-boundary comparison

Capability / trust property	CT / RFC 9162	Key transparency	TLSNotary / endpoint proof	DIYAR (status)
Signed-origin assertion	Not by itself	Not by itself	Web-server transcript under notary assumptions	A1/A2 signed, A2− designed
Append-only history	Yes	Yes	Not inherently	Daily root log implemented locally
Split-view mitigation	Witnesses / gossip	Auditors / witnesses	Not its function	Public-chain checkpoint designed
Non-membership for keyed field	Not generally	Yes	No	Sparse Merkle implemented
Valid time separate from record time	Not native	Not native	No	Bitemporal query implemented
Explicit per-field residual trust grade	No	No	No	7-grade model implemented in prototype paths
Replay derived reports	No	No	No	Published-reports replay implemented
Third-party proof of a private endpoint	No	No	Yes with residual assumptions	A2− is an integration, not invented

Evidence: whitepaper Section V (Related work) and references [18], [21] and [22]; RFC 9162 and the CONIKS paper. Status labels refer to implementation in DIYAR-ref, not provider endorsements.

H.3 Certificate Transparency and witnessed logs

Certificate Transparency (RFC 9162) provides verifiable append-only logs, inclusion proofs and consistency checks. Public monitoring and independent co-signing witnesses reduce the scope of undetected split-view attacks, but a log alone does not assert that the physical meter was installed at the property or that derived climate-performance values were calculated from a particular historical model. A witnessed log with independent data escrow and reproducible computation is a credible alternative to much of DIYAR’s infrastructure.

DIYAR adopts a signed sequence of committed roots and adds field-level grades, input/decision provenance, a separately replayable derivation and bitemporal context. Its planned public blockchain checkpoint seeks permissionless, independently discoverable anchoring without relying solely on a designated consortium member to keep an honest shared view. **That incremental benefit is a design hypothesis, not a demonstrated superiority**, because the Ethereum contract is not deployed: its non-equivocation constraint has been tested only on a local EVM (Annex B).

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H.4 Key transparency is the architectural ancestor

CONIKS-style key transparency and later auditable-key-directory mechanisms map stable names to authenticated values and commit successive epochs. They can support non-membership or verifiable map semantics and already counter certain equivocation patterns. DIYAR's per-field sparse Merkle commitment therefore does not establish novelty by itself. The proposed difference is **domain composition**: a field's origin route and residual assumption, the time the fact was true versus when the system learned it, and reproducible output histories tied to committed models and corrections.

A custom application built on a key-transparency service could incorporate many of these capabilities. DIYAR's contribution is a specified integration and its evaluated boundary conditions in building sustainability, not an impossibility theorem about alternative architectures.

H.5 TLSNotary / zkTLS is an input adapter, not a rival ledger

The A2− grade adopts endpoint-proof concepts from TLSNotary: verifying that certain web-origin content was received under a notary/web-PKI trust model. Such a proof does not by itself create durable building histories, solve physical meter identity, calibrate an anomaly detector, or adjudicate a valuer's claim. A notarised feed may still depend on a colluding notary, the server's correctness, and the web PKI linkage between a domain and the intended utility or other authoritative entity.

The design recommends an MPC-TLS route; proxy and trusted-execution alternatives carry different trust assumptions and depend on protocol and version support. A multi-notary threshold and slashing are future governance proposals. The reference repository does not implement endpoint transcript notarisation, and "publicly verifiable" does not mean "trustless". In a direct utility-data pilot the endpoint-proof route may not be needed at all.

H.6 Threat comparison: operator failure, withholding and equivocation

Case 1: operator goes offline. A single central database becomes unavailable if its only holder disappears. A witnessed log plus independent snapshots and published replay code can preserve independent verification without a public chain; DIYAR's proposed diversified RS custody and public anchor aim for a similar but more widely checkpointed recovery path. Today, the local replay of published files is demonstrated; multi-organisation recovery is not.

Case 2: operator sends conflicting reports. A private signing key can sign 2 internally plausible histories. An append-only log with independent witnesses can detect or constrain split views under an honest-witness assumption. DIYAR's state-root continuity rule prevents 2 competing day roots from being accepted in the same contract state: on a local EVM the competing write reverts (Annex B), but the contract is not deployed. An **event-only transaction** is not equivalent to a state-enforced previous-root rule. The cost model (Annex D) prices the state contract at its measured gas.

Case 3: operator hides a required field. A sparse Merkle non-membership proof establishes absence **relative to a particular root**. Unless mandatory field names and the enrolment population were independently committed before the reporting period, absence cannot prove a missing legal obligation or intent. DIYAR's mandatory-input register is a design control, not implemented population reconciliation. DIYAR-ref can prove a field's absence against a signed root (replay check 7), but its reports do not attach that proof to every missing field.

Case 4: a meter signs false reality. Every listed cryptographic mechanism, including a public chain, can preserve a false signed reading faithfully. Device-location registries, authenticated utility topology checks, random independent inspection and explicit downgraded provenance grades are the correct residual controls. This is the limitation that motivates DIYAR's graded evidence layer rather than a "blockchain makes data true" claim.

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H.7 Why any blockchain, and when not to use one

If all participants accept the same long-lived institution and witnesses can jointly retain independent snapshots, a signed and auditable database or witnessed transparency log may meet the objective with fewer moving parts. A public-chain checkpoint is most defensible only when the evidence must survive changes in operators, lenders, jurisdictions and institutional trust over decades, and when a permissionless external history materially reduces the risk of unilateral rewriting or selective preservation.

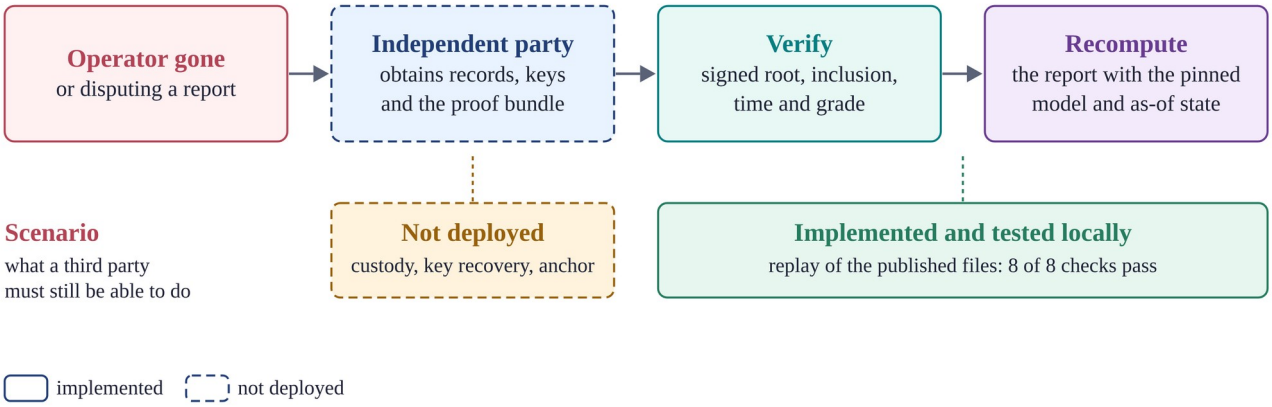


Fig. 1 Platform-disappearance counterfactual. The locally implemented replay test is distinct from the proposed independent storage, key recovery and public-chain anchoring systems.

The chain does **not** improve raw measurement accuracy, remove lawful data-access requirements, generate a secondary market, or deliver automatic legal recognition. DIYAR stores commitments on the proposed chain, not tenant raw data. A cost-benefit analysis must include the contract's gas (measured on a local EVM in Annex D, to be confirmed by mainnet receipts), public-header proof preservation, blockchain dependency and data-removal obligations. The current reference validates the internal proof spine and the contract's rules on a local EVM, but not a public checkpoint.

H.8 Differentiation that can survive adversarial review

The strongest defensible value claim is narrow: **an evidence claim carries its own route-of-arrival grade, residual trust assumptions, typed revision history, reproducible derivation and a portable verification receipt**. A downstream valuer or lender can see which measurements were directly signed, which fields depend on an interested party, what changed after publication, which statistical conditions failed, and whether the same report can be reconstructed years later without trusting the current operator.

The counterfactual to invite is: "What remains if the platform disappears, the meter is misbound, or the peer set becomes dishonest?" The prototype answers the replay question on published files, binds each meter through a signed installation certificate and quantifies a statistical limit. It leaves live registry access, distributed custody, approved keys, enforceable notary thresholds and public-chain anchoring as pilot prerequisites. That calibrated boundary, rather than a novelty claim, is the project's contribution.

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